

# PlatEMO

## 进化多目标优化平台

用户手册 4.17

生物智能与知识发现（BIMK）研究所

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非常感谢使用由安徽大学生物智能与知识发现（BIMK）研究所开发的进化多目标优化平台 PlatEMO。本平台是一个开源免费的代码库，仅供教学与科研使用，不得用于商业用途。本平台中的代码基于作者对论文的理解编写而成，作者不对用户因使用代码产生的任何后果负责。包含利用本平台产生的数据的论文应在正文中声明对 PlatEMO 的使用，并引用以下参考文献之一：

[1] Ye Tian, Weijian Zhu, Xingyi Zhang, and Yaochu Jin, “A practical tutorial on solving optimization problems via PlatEMO,” Neurocomputing, 2023, 518: 190-205.

[2] Ye Tian, Ran Cheng, Xingyi Zhang, and Yaochu Jin, “PlatEMO: A MATLAB platform for evolutionary multi-objective optimization [educational forum],” IEEE Computational Intelligence Magazine, 2017, 12(4): 73-87.

如有任何意见或建议，欢迎联系 field910921@gmail.com（田野）。如想将您的代码添加进 PlatEMO 中并公开，也欢迎联系 field910921@gmail.com。您可以在 GitHub 上获取 PlatEMO 的最新版本。

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# 一 快速入门

软件要求：MATLAB R2018a 或以上（不使用 PlatEMO 图形界面）或
MATLAB R2020b 或以上（使用 PlatEMO 图形界面）及
并行计算工具箱 和
统计与机器学习工具箱

PlatEMO 是一个用于求解优化问题的开源平台，它的输入是一个优化问题，输出是在该优化问题上得到的最优解。一个优化问题满足以下定义：

$$\begin{aligned}\min_{\mathbf{x}} \quad & \mathbf{f}(\mathbf{x}) = \left(f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x})\right) \\ \text{s.t.} \quad & \mathbf{x} = (x_1, x_2, \dots, x_D) \in \Omega \\ & g_1(\mathbf{x}), g_2(\mathbf{x}), \dots, g_K(\mathbf{x}) \le 0\end{aligned}$$

其中 $\mathbf{x}$ 表示该问题的一个解或决策向量，它由 $D$ 个决策变量 $x_i$ 组成，其中每个决策变量可能被限制为实数、整数或二进制数等。$\Omega$ 表示该问题的搜索空间，它由下界 $l_1, l_2, \dots, l_D$ 和上界 $u_1, u_2, \dots, u_D$ 构成，即任意决策变量始终满足 $l_i \le x_i \le u_i$。$f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x})$ 表示该解的 $M$ 个目标函数值，$g_1(\mathbf{x}), g_2(\mathbf{x}), \dots, g_K(\mathbf{x})$ 表示该解的 $K$ 个约束违反值。

为了定义一个优化问题，用户至少需要输入以下内容：

- 每个决策变量的编码方式（实数、整数或二进制数等）；
- 决策变量的下界 $l_1, l_2, \dots, l_D$ 和上界 $u_1, u_2, \dots, u_D$；
- 至少一个目标函数 $f_1(\mathbf{x})$。

为了更精准地定义问题，用户还能输入以下内容：

- 多个目标函数 $f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x})$；
- 多个约束函数 $g_1(\mathbf{x}), g_2(\mathbf{x}), \dots, g_K(\mathbf{x})$；
- 解的初始化函数；
- 无效解的修复函数；
- 解的评价函数；
- 目标和约束的梯度函数；

- 各函数计算中使用到的数据（一个任意类型的常量）。

以上函数均指的是代码函数而非数学函数，即它需要有符合规定的输入和输出，但不需要有显式的数学表达式。此外，用户还能定义与优化算法相关的内容，通过选择合适的算法和参数设置以提升优化效果。

在 MATLAB 中，用户可以用以下三种方式运行主函数文件 `platemo.m`：

1) 带参数调用主函数：

```
platemo('problem',@SOP_F1,'algorithm',@GA);
```

可以利用指定的算法来求解指定的测试问题并设置参数，优化结果可以被显示在窗口中、保存在文件中或作为函数返回值（参阅*求解测试问题*章节）。

2) 带参数调用主函数：

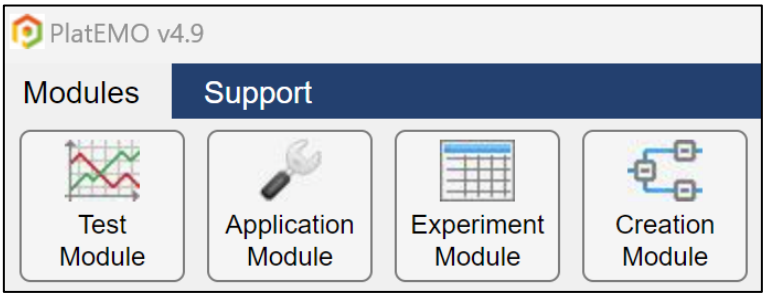
```
f1 = @(x)sum(x);
g1 = @(x)1-sum(x);
platemo('objFcn',f1,'conFcn',g1,'algorithm',@GA);
```

可以利用指定的算法来求解自定义的问题（参阅*求解自定义问题*章节）。

3) 不带参数调用主函数：

```
platemo();
```

可以弹出一个带有四个模块的图形界面，其中测试模块用于可视化地研究单个算法在单个问题上的性能（参阅*测试模块*章节），应用模块用于求解自定义问题（参阅*应用模块*章节），实验模块用于统计分析多个算法在多个问题上的性能（参阅*实验模块*章节），创造模块用于零代码创建全新的 NeuroEA 算法（参阅*创造模块*章节）。



# 二 通过命令行使用 PlatEMO

## 1. 求解测试问题

用户可以以如下形式带参数调用主函数 platemo() 来求解测试问题：

```
platemo('Name1',Value1,'Name2',Value2,'Name3',Value3);
```

其中所有可接受的参数列举如下：

| 参数名 | 数据类型 | 默认值 | 描述 |
|---|---|---|---|
| 'algorithm' | 函数句柄或单元数组 | 不定 | 要运行的算法类 |
| 'problem' | 函数句柄或单元数组 | 不定 | 要求解的问题类 |
| 'N' | 正整数 | 100 | 种群大小 |
| 'M' | 正整数 | 不定 | 问题的目标数 |
| 'D' | 正整数 | 不定 | 问题的变量数 |
| 'maxFE' | 正整数 | 10000 | 最大评价次数 |
| 'maxRuntime' | 正数 | inf | 最大运行时间 |
| 'save' | 整数 | -10 | 保存的种群数 |
| 'run' | 正整数 | [] | 当前运行的编号 |
| 'metName' | 字符串或单元数组 | {} | 要计算的指标名称 |
| 'outputFcn' | 函数句柄 | @DefaultOutput | 每代开始前调用的函数<br>输入一：ALGORITHM 对象<br>输入二：PROBLEM 对象<br>输出：无 |

- 'algorithm'表示待运行的算法，它的值可以是一个算法类的句柄，例如@GA。它的值还可以是形如{@GA,p1,p2,…}的单元数组，其中 p1,p2,… 指定了该算法中的参数值。例如以下代码用算法@GA 求解默认问题，并设置了该算法中的参数值：

```
platemo('algorithm',{@GA,1,30,1,30});
```

- 'problem'表示待求解的测试问题，它的值可以是一个问题类的句柄，例

如@SOP_F1。它的值还可以是形如{@SOP_F1,p1,p2,…}的单元数组，其中 p1,p2,… 指定了该问题中的参数值。例如以下代码用默认算法求解问题@WFG1，并设置了该问题中的参数值：

```
platemo('problem',{@WFG1,20});
```

- 'N'表示算法使用的种群的大小，它通常等于最终输出的解的个数。例如以下代码用算法@GA 求解问题@SOP_F1，并设置种群大小为 50：

```
platemo('algorithm',@GA,'problem',@SOP_F1,'N',50);
```

- 'M'表示问题的目标个数，它仅对一些多目标测试问题生效。例如以下代码用算法@NSGAII 求解具有 5 个目标的@DTLZ2 问题：

```
platemo('algorithm',@NSGAII,'problem',@DTLZ2,'M',5);
```

- 'D'表示问题的变量个数，它仅对一些测试问题生效。例如以下代码用算法@GA 求解具有 100 个变量的@SOP_F1 问题：

```
platemo('algorithm',@GA,'problem',@SOP_F1,'D',100);
```

- 'maxFE'表示算法可用的最大评价次数，它通常等于种群大小乘以迭代次数。例如以下代码设置算法@GA 的最大评价次数为 20000：

```
platemo('algorithm',@GA,'problem',@SOP_F1,'maxFE',20000);
```

- 'maxRuntime'表示算法可用的最大运行时间，单位为秒。当'maxRuntime'等于默认值 inf 时，算法将在'maxFE'次评价次数后停止；否则，算法将在'maxRuntime'秒后停止。例如以下代码设置算法@GA 的最大运行时间为 10 秒：

```
platemo('algorithm',@GA,'problem',@SOP_F1,'maxRuntime',10);
```

- 'save'表示保存的种群数，该值大于零时优化结果将被保存在文件中，该值小于零时优化结果将被显示在窗口中（参阅获取运行结果章节）。
- 'run'表示当前运行的编号，它附加在保存文件名的末尾，使相同算法在相同问题上的多次运行结果对应的文件名不同（参阅获取运行结果章节）。
- 'metName'表示要计算的指标名称，它可以是一个字符串（单个指标）或一个单元数组（多个指标）。保存的种群会被计算指定的指标值，并保存在文件或显示在窗口中（参阅获取运行结果章节）。
- 'outputFcn'表示算法每代开始前调用的函数。该函数必须有两个输入和

零个输出，其中第一个输入是当前的 ALGORITHM 对象、第二个输入是当前的 PROBLEM 对象。默认的'outputFcn'会根据'save'的值来保存或显示优化结果。

注意以上每个参数均有一个默认值，用户可以在调用时省略任意参数。

## 2. 求解自定义问题

当不指定参数'problem'时，用户可以通过指定以下参数来自定义问题：

| 参数名 | 数据类型 | 默认值 | 描述 |
|---|---|---|---|
| 'objFcn' | 函数句柄、矩阵或单元数组 | {} | 问题的目标函数；所有目标函数均被最小化<br>输入：一个决策向量<br>输出：目标值（标量） |
| 'encoding' | 标量或行向量 | 1 | 每个变量的编码方式 |
| 'lower' | 标量或行向量 | 0 | 每个变量的下界 |
| 'upper' | 标量或行向量 | 1 | 每个变量的上界 |
| 'conFcn' | 函数句柄、矩阵或单元数组 | {} | 问题的约束函数；当且仅当约束违反值小于等于零时，该约束被满足<br>输入：一个决策向量<br>输出：约束违反值（标量） |
| 'decFcn' | 函数句柄 | {} | 无效解修复函数<br>输入：一个决策向量<br>输出：修复后的决策向量 |
| 'evalFcn' | 函数句柄 | {} | 解的评价函数<br>输入：一个决策向量<br>输出一：修复后的决策向量<br>输出二：所有目标值（向量）<br>输出三：所有约束违反值（向量） |
| 'initFcn' | 函数句柄 | {} | 种群初始化函数<br>输入：种群大小<br>输出：种群的决策向量构成的矩阵 |
| 'gradFcn' | 函数句柄 | {} | 目标和约束的梯度函数<br>输入：一个决策向量<br>输出一：目标雅可比矩阵<br>输出二：约束雅可比矩阵 |
| 'data' | 任意 | {} | 问题的数据 |
| 'once' | 逻辑 | 0 | 是否支持同时评价多个解 |

- 'objFcn'表示问题的目标函数，它的值可以是一个函数句柄（单目标）、矩阵（自动拟合出函数）或一个单元数组（多目标）。每个目标函数必须有一个输入和一个输出，其中输入是一个决策向量、输出是目标值。所有目标函数均被最小化。例如以下代码利用默认算法求解一个含有六个实数变量的双目标优化问题：

```
f1 = @(x)x(1)+sum(x(2:end));
f2 = @(x)sqrt(1-x(1)^2)+sum(x(2:end));
platemo('objFcn',{f1,f2},'D',6);
```

其中第一个目标为 $x_1+\sum_{i=2}^{D}x_i$、第二个目标为 $\sqrt{1-x_1^2}+\sum_{i=2}^{D}x_i$。若一个目标函数是矩阵，则高斯过程回归会利用该矩阵自动拟合出一个函数，其中矩阵的每行表示一个样本、每列表示一个变量（除最后一列）或函数值（最后一列）。例如以下代码求解相同的问题，但目标函数是根据矩阵自动拟合出来的：

```
x = rand(50,6);
y1 = x(:,1)+sum(x(:,2:end),2);
y2 = sqrt(1-x(:,1).^2)+sum(x(:,2:end),2);
platemo('objFcn',{[x,y1],[x,y2]},'D',6);
```

- 'encoding'表示每个变量的编码方式，它的值可以是一个标量或行向量，且每维的值可以为 1（实数）、2（整数）、3（标签）、4（二进制数）或 5（序列编号）。算法针对不同的编码方式可能使用不同的算子来产生解。例如以下代码指定三个实数变量、两个整数变量以及一个二进制变量：

```
f1 = @(x)x(1)+sum(x(2:end));
f2 = @(x)sqrt(1-x(1)^2)+sum(x(2:end));
platemo('objFcn',{f1,f2},'encoding',[1,1,1,2,2,4]);
```

问题的变量数 $D$ 将根据'encoding'的长度自动确定。

- 'lower'和'upper'分别表示每个变量的下界和上界，它们的值可以是标量或行向量，且每维的值必须为实数。'lower'和'upper'的长度必须与'encoding'相同。例如以下代码指定搜索空间为 $[0,1]\times[0,9]^5$：

```
f1 = @(x)x(1)+sum(x(2:end));
f2 = @(x)sqrt(1-x(1)^2)+sum(x(2:end));
platemo('objFcn',{f1,f2},'encoding',[1,1,1,2,2,4],...
'lower',0,'upper',[1,9,9,9,9,9]);
```

- 'conFcn'表示问题的约束函数，它的值可以是一个函数句柄（单约束）、矩阵（自动拟合出函数）或一个单元数组（多约束）。每个约束函数必须有一个输入和一个输出，其中输入是一个决策向量、输出是约束违反值。当且仅当约束违反值小于等于零时，该约束被满足。例如以下代码利用默认算法求解一个双目标优化问题：

```
f1 = @(x)x(1)+sum(x(2:end));
f2 = @(x)sqrt(1-x(1)^2)+sum(x(2:end));
g1 = @(x)1-sum(x(2:end));
platemo('objFcn',{f1,f2},'encoding',[1,1,1,2,2,4],...
'conFcn',g1,'lower',0,'upper',[1,9,9,9,9,9]);
```

并添加约束函数 $\sum_{i=2}^{6} x_i \geq 1$。注意，等式约束必须转换为不等式约束来处理，详细方法可参阅该论文的 3.2 节。若一个约束函数是矩阵，则高斯过程回归会利用该矩阵自动拟合出一个函数，其中矩阵的每行表示一个样本、每列表示一个变量（除最后一列）或函数值（最后一列）。例如以下代码求解相同的问题，但约束函数是根据矩阵自动拟合出来的：

```
f1 = @(x)x(1)+sum(x(2:end));
f2 = @(x)sqrt(1-x(1)^2)+sum(x(2:end));
x = rand(50,6);
y = 1-sum(x(:,2:end),2);
platemo('objFcn',{f1,f2},'encoding',[1,1,1,2,2,4],...
'conFcn',[x,y],'lower',0,'upper',[1,9,9,9,9,9]);
```

- 'decFcn'表示问题的无效解修复函数，它的值必须是一个函数句柄。该函数必须有一个输入和一个输出，其中输入是一个决策向量、输出是修复后的决策向量。默认的'decFcn'将所有解的范围限定在'lower'和'upper'之间，而以下代码定义了一个新的'decFcn'限制 $x_1$ 为 0.1 的倍数：

```
f1 = @(x)x(1)+sum(x(2:end));
f2 = @(x)sqrt(1-x(1)^2)+sum(x(2:end));
g1 = @(x)1-sum(x(2:end));
h = @(x)[round(x(1)/0.1)*0.1,x(2:end)];
platemo('objFcn',{f1,f2},'encoding',[1,1,1,2,2,4],...
'conFcn',g1,'decFcn',h,'lower',0,'upper',[1,9,9,9,9,9]);
```

- 'evalFcn'表示解的评价函数，它的值必须是一个函数句柄。该函数必须有一个输入和三个输出，其中输入是一个决策向量、第一个输出是修复后的决策向量、第二个输出是目标值向量、第三个输出是约束违反值向量。默认

的'evalFcn'通过依次调用'decFcn'、'objFcn'和'conFcn'来评价解，而以下代码定义了一个新的'evalFcn'来同时进行解的修复、目标计算和约束计算：

```
function [x,f,g] = Eval(x)
    x = [round(x(1)/0.1)*0.1,x(2:end)];
    x = max(0,min([1,9,9,9,9,9],x));
    f(1) = x(1)+sum(x(2:end));
    f(2) = sqrt(1-x(1)^2)+sum(x(2:end));
    g = 1-sum(x(2:end));
end
```

接着，以下代码通过仅指定评价函数定义了相同的问题：

```
platemo('evalFcn',@Eval,'encoding',[1,1,1,2,2,4],...
'lower',0,'upper',[1,9,9,9,9,9]);
```

- 'initFcn'表示种群初始化函数，它的值必须是一个函数句柄。该函数必须有一个输入和一个输出，其中输入是种群大小、输出是种群的决策向量构成的矩阵。默认的'initFcn'在整个搜索空间内随机产生初始解，而以下代码定义了一个新的'initFcn'以加速收敛：

```
q = @(N)rand(N,6);
platemo('evalFcn',@Eval,'encoding',[1,1,1,2,2,4],...
'initFcn',q,'lower',0,'upper',[1,9,9,9,9,9]);
```

- 'gradFcn'表示目标和约束的梯度函数，它的值必须是一个函数句柄。该函数必须有一个输入和两个输出，其中输入是一个决策向量、第一个输出是目标雅可比矩阵、第二个输出是约束雅可比矩阵。默认的梯度函数通过有限差分来估计梯度，而以下代码定义了一个新的'gradFcn'以加速收敛：

```
function [oGrad,cGrad] = Grad(x)
    oGrad = [0,x(2:end);0,x(2:end)];
    cGrad = [0,x(2:end)-1/5];
end
```

接着，以下代码通过指定梯度函数来更好地求解问题：

```
platemo('evalFcn',@Eval,'encoding',[1,1,1,2,2,4],...
'gradFcn',@Grad,'lower',0,'upper',[1,9,9,9,9,9]);
```

注意仅有少量算法会使用梯度函数。

- 'data'表示问题的数据，它可以是任意类型的常量。当指定'data'后，以

上所有函数必须增加一个输入参数来接收'data'。例如以下代码求解一个旋转的单目标优化问题：

```
d = rand(RandStream('mlfg6331_64','Seed',28),10)*2-1;
[d,~] = qr(d);
f1 = @(x,d)sum((x*d-0.5).^2);
platemo('objFcn',f1,'encoding',ones(1,10),'data',d);
```

- 'once'表示是否可以同时评价多个解，它是默认值为零的逻辑变量。当指定'once'的值为 1 后，'evalFcn'、'decFcn'、'objFcn'和'conFcn'的输入可以为多个决策向量，即同时评价多个解。在函数中使用矩阵运算或并行计算来支持同时评价多个解，可以显著提升求解效率。例如以下代码将目标函数改写为矩阵运算：

```
d = rand(RandStream('mlfg6331_64','Seed',28),10)*2-1;
[d,~] = qr(d);
f1 = @(x,d)sum((x*d-0.5).^2,2);
platemo('objFcn',f1,'encoding',ones(1,10),'data',d,'once',1);
```

除以上定义问题的方式之外，用户还能创建一个自定义问题对象并创建算法对象予以求解。例如以下代码利用算法@GA 和算法@DE 求解相同的问题：

```
d = rand(RandStream('mlfg6331_64','Seed',28),10)*2-1;
[d,~] = qr(d);
f1 = @(x,d)sum((x*d-0.5).^2);
PRO = UserProblem('objFcn',f1,'encoding',ones(1,10),'data',d);
ALG1 = GA();
ALG2 = DE();
ALG1.Solve(PRO);
ALG2.Solve(PRO);
```

### 3. 获取运行结果

算法运行结束后得到的种群可以被显示在窗口中、保存在文件中或作为函数返回值。若按以下方式调用主函数：

```
[Dec,Obj,Con] = platemo('Name1',Value1,'Name2',Value2);
```

则最终种群会被返回，其中 Dec 表示种群的决策向量构成的矩阵、Obj 表示种群的目标值构成的矩阵、Con 表示种群的约束违反值构成的矩阵。若按以下方式调用主函数：

```
platemo('save',Value);
```

则当 Value 的值为负整数时（默认情况），得到的种群会被显示在窗口中，用户可以在窗口中的 Data source 菜单选择要显示的内容。当 Value 的值为正整数时，得到的种群会被保存在名为 PlatEMO\Data\alg\alg_pro_M_D_run.mat 的 MAT 文件中，其中 alg 表示算法名、pro 表示问题名、M 表示目标数、D 表示变量数、run 是一个自动确定的正整数以保证不和已有文件重名。同时，可按以下方式主动指定 run 的值：

```
parfor i = 1 : 100
    platemo('save',Value,'run',i);
end
```

则 run 的值会被指定为 1 到 100。在并行多次运行时，主动指定 run 的值可以避免文件编号混乱或缺失。

每个保存的数据文件存储一个单元数组 result 和一个结构体 metric，其中 result 保存得到的种群、metric 保存指标值。算法的整个优化过程被等分为 Value 块，其中 result 的第一列存储每块最后一代时所消耗的评价次数、result 的第二列存储每块最后一代时的种群、metric 存储所有种群的指标值。

```
result =
  6×2 cell array
    {[ 1600]}    {1×100 SOLUTION}
    {[ 3300]}    {1×100 SOLUTION}
    {[ 5000]}    {1×100 SOLUTION}
    {[ 6600]}    {1×100 SOLUTION}
    {[ 8300]}    {1×100 SOLUTION}
    {[10000]}    {1×100 SOLUTION}
```

```
metric =
  struct with fields:

    runtime: 0.2267
        IGD: [6×1 double]
         HV: [6×1 double]
```

可以通过参数'metName'来指定要计算的指标，例如以下代码用算法@NSGAII 求解@DTLZ2 问题，并计算 IGD 和 HV 指标值保存在文件中：

```
platemo('algorithm',@NSGAII,'problem',@DTLZ2,...
'save',6,'metName',{'IGD','HV'});
```

其中'IGD'和'HV'为要计算的指标名（参阅*指标函数*章节）。特别地，IGD 和 HV 是多目标优化中最常用的性能指标，它们的适用范围和参考点定义方法参阅该论文的 5.3 节。以上操作均由默认的输出函数@DefaultOutput 实现，用户可以通过指定'outputFcn'的值为其它函数来实现自定义的结果展示或保存方式。此外，可按以下方式计算单个种群的指标值：

```
% 在执行以下代码之前需先载入 result
pro = DTLZ2();
pro.CalMetric('IGD',result{end});
```

同时，图形界面的实验模块可以自动计算种群的指标值并存储到文件中。

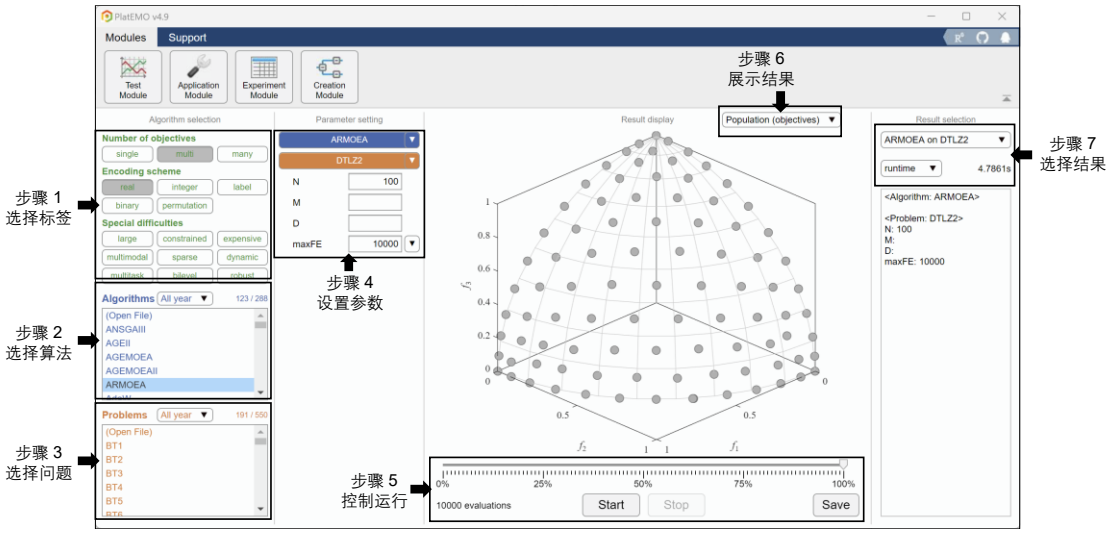
# 三　通过图形界面使用 PlatEMO

## 1. 测试模块

用户可以通过无参数调用主函数 `platemo()` 来使用 PlatEMO 的图形界面：

```
platemo();
```

图形界面的测试模块会被首先显示，它用于可视化地研究单个算法在单个问题上的性能。

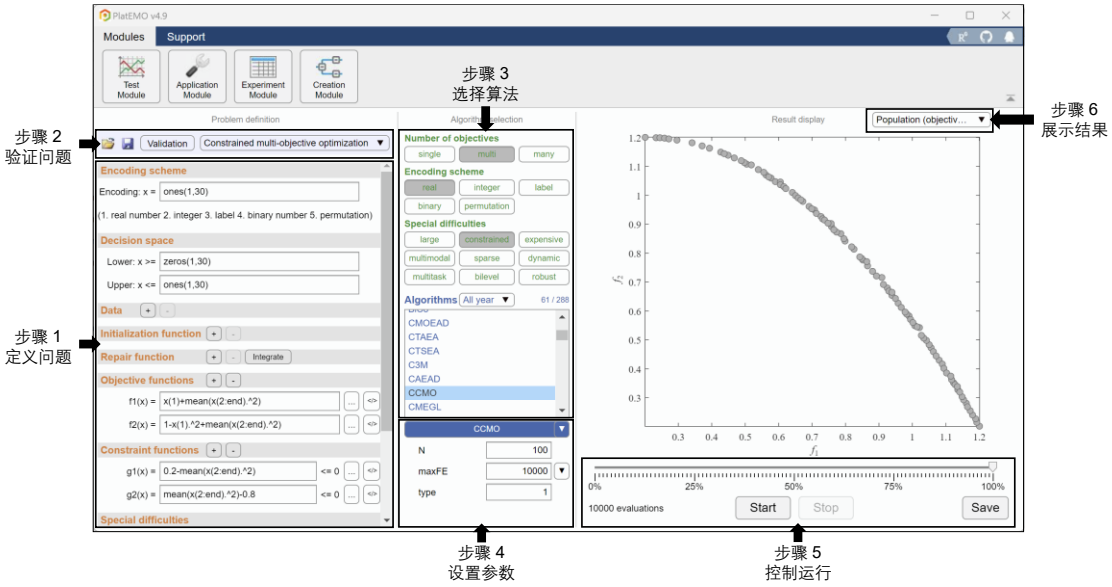



在该模块中，用户能用以下步骤研究单个算法在单个问题上的性能：

- 步骤 1：选择多个标签确定问题类型（参阅*算法、问题和指标的标签*章节）。
- 步骤 2：在列表中选择一个算法。
- 步骤 3：在列表中选择一个问题。
- 步骤 4：设置算法和问题的参数。不同算法和问题可能有不同的参数，在参数上悬停可查看具体说明。
- 步骤 5：开始、暂停、停止或回退算法的运行；保存当前结果到文件。当前结果可被保存为一个 $N$ 行 $D+M+K$ 列的矩阵，$N$ 表示解的个数，$D$ 表示决策变量个数，$M$ 表示目标个数，$K$ 表示约束个数。
- 步骤 6：选择要显示的数据，例如当前种群的目标值、变量值和各指标值。
- 步骤 7：选择要显示的历史运行结果。

## 2. 应用模块

用户可以通过图形界面中的菜单切换至应用模块，它用于求解自定义问题。

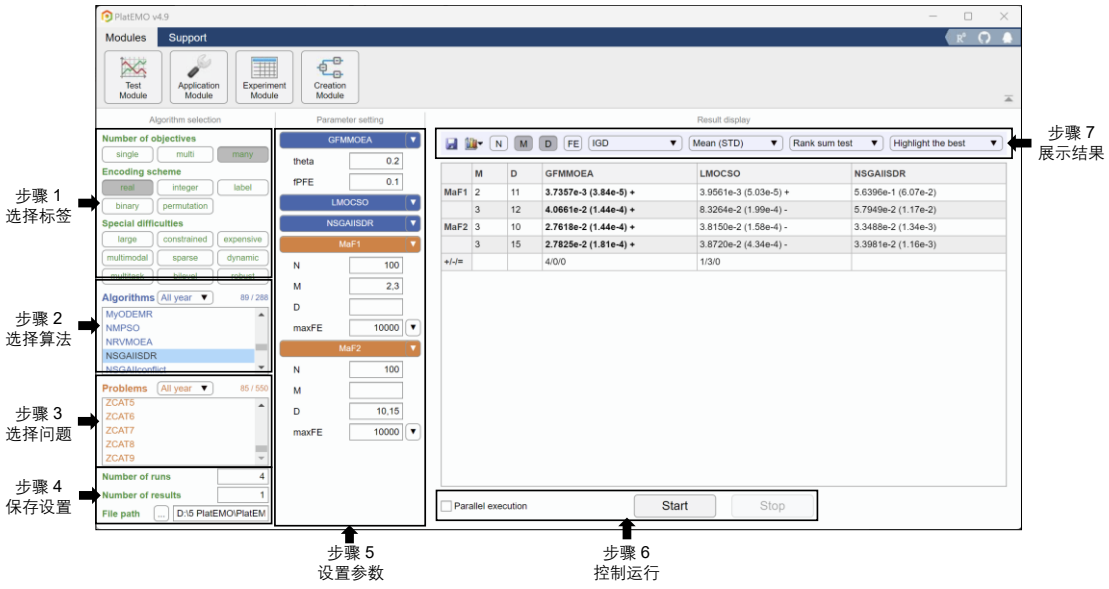



在该模块中，用户能用以下步骤求解自定义问题：

- 步骤 1：定义一个问题，定义的内容与求解自定义问题相同，其中 Encoding scheme 对应`'encoding'`，Decision space 对应`'lower'`和`'upper'`，Data 对应`'data'`，Initialization function 对应`'initFcn'`，Repair function 对应`'decFcn'`，Objective functions 对应`'objFcn'`，Constraint functions 对应`'conFcn'`，Evaluation function 对应`'evalFcn'`。
- 步骤 2：保存或载入问题；检测问题定义的合法性；选择一个问题模板。保存的问题可在其它模块中打开并求解。
- 步骤 3：在列表中选择一个算法。标签会根据问题定义自动确定(参阅算法、问题和指标的标签章节)。
- 步骤 4：设置算法的参数。不同算法可能有不同的参数，在参数上悬停可查看具体说明。
- 步骤 5：开始、暂停、停止或回退算法的运行；保存当前结果到文件。当前结果可被保存为一个 $N$ 行 $D+M+K$ 列的矩阵，$N$ 表示解的个数，$D$ 表示决策变量个数，$M$ 表示目标个数，$K$ 表示约束个数。
- 步骤 6：选择要显示的数据，例如种群的目标值、变量值和各指标值。

## 3. 实验模块

用户可以通过图形界面中的菜单切换至实验模块，它用于统计分析多个算法在多个问题上的性能。该模块中所有优化结果将被保存至 MAT 文件（参见获取运行结果章节），如文件存在则会直接读取而不运行算法。

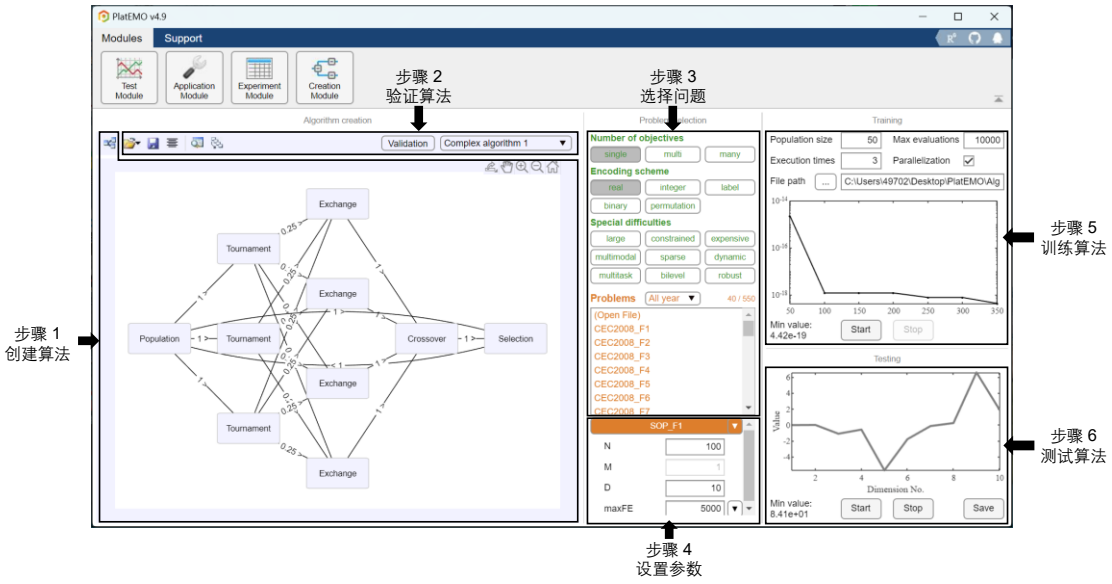



在该模块中，用户能用以下步骤比较多个算法在多个问题上的性能：

- 步骤 1：选择多个标签确定问题类型(参阅算法、问题和指标的标签章节)。
- 步骤 2：在列表中选择多个算法。
- 步骤 3：在列表中选择多个问题。
- 步骤 4：设置实验重复次数、每次保存的种群个数及保存的文件路径（参阅获取运行结果章节)。
- 步骤 5：设置算法和问题的参数。不同算法和问题可能有不同的参数，在参数上悬停可查看具体说明。此处问题的参数可以设置为向量，这使得同一个问题可以产生多个不同的测试实例。
- 步骤 6：开始或停止实验的运行；选择串行（单 CPU）或并行（多 CPU）运行实验。
- 步骤 7：选择要显示的指标值；选择要执行的统计分析；保存表格到文件；将选中的多个单元格的数据显示在图窗中。

## 4. 创造模块

用户可以通过图形界面中的菜单切换至创造模块，它用于创建全新的 NeuroEA 算法，并在指定问题上训练它。关于 NeuroEA 算法的细节可参阅该论文，以无界面的方式创建 NeuroEA 算法的方法可参阅创建 NeuroEA 算法章节。




在该模块中，用户能用以下步骤创建并训练算法：

- 步骤 1：通过点击按钮来添加模块，通过点击两个模块来添加连接，通过拖动模块和连接来改变布局。模块包含种群模块、算子模块和选择模块，每个模块有一些预设的超参数和一些待训练的参数；连接表示模块间解的传递方向和比例。一个算法视为一个以模块为节点、以连接为边的有权有向循环图，其中第一个节点必须为种群模块、算法至少包含一个算子模块节点、所有节点必须有前驱和后继节点、所有节点必须互相可达、所有环中必须包含至少一个种群模块节点。
- 步骤 2：保存或载入算法或模块；生成算法代码；改变显示样式；自动排列模块；检测算法的合法性；选择一个算法模板。算法训练完成后，可生成算法代码并在其它模块使用。
- 步骤 3：选择多个标签确定问题类型(参阅算法、问题和指标的标签章节)；在列表中选择一个问题。
- 步骤 4：设置问题的参数。不同问题可能有不同的参数，在参数上悬停可查看具体说明。
- 步骤 5：在选择的问题上训练算法中所有模块的参数。这个过程可能较慢，

较大的模块数目、问题变量数目、种群大小和评价次数可能耗费数天。

- 步骤 6：在选择的问题上测试训练后的算法的性能。

## 5. 算法、问题和指标的标签

每个算法、测试问题和指标需要被添加上标签，这些标签以注释的形式添加在主函数代码的第二行。例如在 PSO.m 代码的开头部分：

```
classdef PSO < ALGORITHM
% <1995> <single> <real/integer> <large/none> <constrained/none>
```

通过多个标签指定了该算法可求解的问题类型。所有的标签列举如下：

| 标签 | 描述 |
|---|---|
| <single> | 单目标优化：问题含有一个目标函数 |
| <multi> | 多目标优化：问题含有两个或三个目标函数 |
| <many> | 超多目标优化：问题含有四个或更多目标函数 |
| <real> | 连续优化：决策变量为实数 |
| <integer> | 整数优化：决策变量为整数 |
| <label> | 标签优化：决策变量为标签 |
| <binary> | 二进制优化：决策变量为二进制数 |
| <permutation> | 序列优化：决策变量构成一个排列 |
| <large> | 大规模优化：问题含有 100 或更多的决策变量 |
| <constrained> | 约束优化：问题含有至少一个约束 |
| <expensive> | 昂贵优化：目标函数的计算非常耗时，即最大评价次数非常小 |
| <multimodal> | 多模优化：存在多个目标值接近但决策向量差异很大的最优解，它们都需要被找到 |
| <sparse> | 稀疏优化：最优解中大部分的决策变量均为零 |
| <dynamic> | 动态优化：目标函数和约束函数随时间变化 |
| <multitask> | 多任务优化：同时优化多个问题，每个问题可能含有多个目标函数和约束函数 |
| <bilevel> | 双层优化：旨在寻找上层问题的可行且最优的解，一个解对于上层问题是可行的当且仅当它是下层问题的最优解 |
| <robust> | 鲁棒优化：目标函数和约束函数受噪声影响，旨在寻找受噪声影响尽可能小且尽可能优的解 |
| <none> | 空标签 |
| <min> | （仅用于指标）该指标值越小表示性能越好 |
| <max> | （仅用于指标）该指标值越大表示性能越好 |

每个算法可能含有多个标签集合，这些集合的笛卡尔积构成该算法可求解的所有的问题类型。例如当标签集合为<single> <real> <constrained/none>时，表示该算法可求解带或不带约束的单目标连续优化问题；若标签集合为<single> <real>，表示该算法只能求解无约束问题；若标签集合为<single> <real> <constrained>，表示该算法只能求解有约束问题；若标签集合为<single> <real/binary>，表示该算法可以求解连续或二进制优化问题。

每个算法、测试问题和指标都需要被添加至少一个标签，否则它将不会在图形界面的列表中出现。当用户在图形界面中选择多个标签后，仅有符合该标签组合的算法、测试问题和指标才会被显示以供选择。标签过滤的具体原理可参阅这里。PlatEMO 中所有算法和测试问题的标签分别参阅*算法列表*和*问题列表*章节。

除此之外，每个算法和测试问题可以被添加一个年份标签如<2024>，这使得图形界面的列表中的算法和测试问题可以按年份过滤。

# 四　扩展 PlatEMO

## 1. 算法类

每个算法需要被定义为 ALGORITHM 类的子类并保存在 PlatEMO\Algorithms 文件夹中。算法类包含的属性与方法如下：

| 属性 | 赋值方式 | 描述 |
|---|---|---|
| parameter | 用户 | 算法的参数 |
| save | 用户 | 每次运行中保存的种群数 |
| run | 用户 | 当前运行的编号 |
| metName | 用户 | 要计算的指标名称 |
| outputFcn | 用户 | 在 NotTerminated() 中调用的函数 |
| pro | Solve() | 当前运行中求解的问题对象 |
| result | NotTerminated() | 当前运行中保存的种群 |
| metric | NotTerminated() | 当前保存的种群的指标值 |
| starttime | NotTerminated() | 用于记录当前运行用时 |
| 方法 | 是否可重定义 | 描述 |
| ALGORITHM | 不可 | 设定由用户指定的属性值<br>输入：形如'Name',Value 的参数设置<br>输出：ALGORITHM 对象 |
| Solve | 不可 | 利用算法求解一个问题<br>输入：PROBLEM 对象<br>输出：无 |
| main | 必须 | 算法的主体部分<br>输入：PROBLEM 对象<br>输出：无 |
| NotTerminated | 不可 | main() 中每次迭代前调用的函数<br>输入：SOLUTION 对象数组，即种群<br>输出：是否达到终止条件（逻辑变量） |
| ParameterSet | 不可 | 根据 parameter 设定算法参数<br>输入：默认的参数设置<br>输出：用户指定的参数设置 |

每个算法需要继承 ALGORITHM 类并重定义方法 main()。例如 GA.m 的代码为：

```
1 classdef GA < ALGORITHM
2 % <1992><single><real/integer/label/binary/permutation><large/none><constrained/none>
3 % Genetic algorithm
```

```
4  % proC ---  1 --- Probability of crossover
5  % disC --- 20 --- Distribution index of crossover
6  % proM ---  1 --- Expectation of the number of mutated variables
7  % disM --- 20 --- Distribution index of mutation
8
9  %------------------------ Reference ----------------------
10 % J. H. Holland, Adaptation in Natural and Artificial
11 % Systems, MIT Press, 1992.
12 %---------------------------------------------------------
13
14     methods
15         function main(Alg,Pro)
16             [proC,disC,proM,disM] = Alg.ParameterSet(1,20,1,20);
17             P = Pro.Initialization();
18             while Alg.NotTerminated(P)
19                 Q = TournamentSelection(2,Pro.N,FitnessSingle(P));
20                 O = OperatorGA(P(Q),{proC,disC,proM,disM});
21                 P = [P,O];
22                 [~,rank] = sort(FitnessSingle(P));
23                 P = P(rank(1:Pro.N));
24             end
25         end
26     end
27 end
```

各行代码的功能如下：

第 1 行：　继承 ALGORITHM 类；

第 2 行：　为算法添加标签（参阅算法、问题和指标的标签章节）；

第 3 行：　算法的全称；

第 4-7 行：参数名 --- 默认值 --- 参数描述，将会显示在图形界面的参数设置列表中；

第 9-12 行：算法的参考文献；

第 15 行：　重定义算法主体流程的方法；

第 16 行：　获取用户指定的参数设置，其中 1,20,1,20 分别表示参数 proC, disC,proM,disM 的默认值。

第 17 行：　调用 PROBLEM 类的方法获得一个初始种群；

第 18 行：　保存当前种群并检查是否达到终止条件；若达到终止条件则通过抛出错误强行终止算法；

第 19 行：　调用公共函数实现基于二元联赛的交配池选择；

第 20 行： 调用公共函数产生子代种群；

第 21 行： 将父子代种群合并；

第 22 行： 调用公共函数计算种群中解的适应度，并依此对解进行排序；

第 23 行： 保留适应度较好的一半解进入下一代。

在以上代码中，函数 ParameterSet() 和 NotTerminated() 是 ALGORITHM 类的方法，函数 Initialization()是 PROBLEM 类的方法，而函数 TournamentSelection()、FitnessSingle()和 OperatorGA()是在 PlatEMO\Algorithms\Utility functions 文件夹中的公共函数。所有可被算法调用的方法及公共函数列举如下，详细的调用方式参阅代码中的注释。此外，函数中用于提升算法效率的技术参阅这里。

| 函数名 | 描述 |
|---|---|
| ALGORITHM.NotTerminated | 算法每代前调用的函数，用于保存当前种群及判断是否终止 |
| ALGORITHM.ParameterSet | 根据用户的输入设定算法参数 |
| PROBLEM.Initialization | 初始化一个种群 |
| PROBLEM.Evaluation | 评价一个种群并产生 SOLUTION 对象数组 |
| CrowdingDistance | 计算解的拥挤距离（仅用于多目标优化） |
| FitnessSingle | 计算解的适应度（仅用于单目标优化） |
| NDSort | 非支配排序（仅用于多目标优化） |
| OperatorDE | 差分进化算子 |
| OperatorFEP | 进化规划算子 |
| OperatorGA | 遗传算子 |
| OperatorGAhalf | 遗传算子（仅返回前一半的子代） |
| OperatorPSO | 粒子群优化算子 |
| RouletteWheelSelection | 轮盘赌选择 |
| TournamentSelection | 联赛选择 |
| UniformPoint | 产生均匀分布的参考点 |

## 2. 问题类

每个问题需要被定义为 PROBLEM 类的子类并保存在 PlatEMO\Problems 文件夹中。问题类包含的属性与方法如下：

| 属性 | 赋值方式 | 描述 |
|---|---|---|
| N | 用户 | 求解该问题的算法的种群大小 |
| M | 用户和 Setting() | 问题的目标数 |
| D | 用户和 Setting() | 问题的变量数 |
| maxFE | 用户 | 求解该问题可使用的最大评价次数 |
| FE | Evaluation() | 当前运行中已消耗的评价次数 |
| maxRuntime | 用户 | 求解该问题可使用的最大运行时间（秒） |
| encoding | Setting() | 每个变量的编码方式 |
| lower | Setting() | 每个变量的下界 |
| upper | Setting() | 每个变量的上界 |
| optimum | GetOptimum() | 问题的最优值，例如目标函数的最小值（单目标优化）和前沿面上一组均匀参考点（多目标优化） |
| PF | GetPF() | 问题的前沿面，例如 1 维曲线（双目标优化）、2 维曲面（三目标优化）和可行区域（约束优化） |
| parameter | 用户 | 问题的参数 |
| 方法 | 是否可重定义 | 描述 |
| PROBLEM | 不可 | 设定由用户指定的属性值<br>输入：形如'Name',Value 的参数设置<br>输出：PROBLEM 对象 |
| Setting | 必须 | 设定默认的属性值<br>输入：无<br>输出：无 |
| Initialization | 可以 | 初始化一个种群<br>输入：种群大小<br>输出：SOLUTION 对象数组，即种群 |
| Evaluation | 可以 | 评价一个种群并产生解对象<br>输入：种群的决策向量构成的矩阵<br>输出：SOLUTION 对象数组，即种群 |
| CalDec | 可以 | 修复一个种群中的无效解<br>输入：种群的决策向量构成的矩阵<br>输出：修复后的决策向量构成的矩阵 |
| CalObj | 必须 | 计算一个种群中解的目标值；所有目标函数均被最小化<br>输入：种群的决策向量构成的矩阵<br>输出：种群的目标值构成的矩阵 |
| CalCon | 可以 | 计算一个种群中解的约束违反值；当且仅当约束 |

| | | |
|---|---|---|
| | | 违反值小于等于零时，约束被满足<br>输入：种群的决策向量构成的矩阵<br>输出：种群的约束违反值构成的矩阵 |
| CalGrad | 可以 | 计算一个解在所有目标和约束上的梯度<br>输入：一个决策向量<br>输出一：目标雅可比矩阵<br>输出二：约束雅可比矩阵 |
| GetOptimum | 可以 | 产生问题的最优值并保存在 optimum 中<br>输入：最优值的个数<br>输出：最优值集合（矩阵） |
| GetPF | 可以 | 产生问题的前沿面并保存在 PF 中<br>输入：无<br>输出：用于绘制前沿面的数据（矩阵或单元数组） |
| CalMetric | 可以 | 计算种群的指标值<br>输入一：指标名<br>输入二：SOLUTION 对象数组，即种群<br>输出：指标值（标量） |
| DrawDec | 可以 | 显示一个种群的决策向量<br>输入：SOLUTION 对象数组，即种群<br>输出：无 |
| DrawObj | 可以 | 显示一个种群的目标向量<br>输入：SOLUTION 对象数组，即种群<br>输出：无 |
| ParameterSet | 不可 | 根据 parameter 设定问题参数<br>输入：默认的参数设置<br>输出：用户指定的参数设置 |

每个算法需要继承 PROBLEM 类并重定义方法 Setting()和 CalObj()。例如 SOP_F1.m 的代码为：

```
classdef SOP_F1 < PROBLEM
% <1999><single><real><expensive/none>
% Sphere function

%------------------------ Reference ----------------------
% X. Yao, Y. Liu, and G. Lin, Evolutionary programming made
% faster, IEEE Transactions on Evolutionary Computation,
% 1999, 3(2): 82-102.
%---------------------------------------------------------

    methods
        function Setting(obj)
```

```
13            obj.M = 1;
14            if isempty(obj.D); obj.D = 30; end
15            obj.lower = zeros(1,obj.D) - 100;
16            obj.upper = zeros(1,obj.D) + 100;
17            obj.encoding = ones(1,obj.D);
18        end
19        function PopObj = CalObj(obj,PopDec)
20            PopObj = sum(PopDec.^2,2);
21        end
22    end
23 end
```

各行代码的功能如下：

第 1 行：　继承 PROBLEM 类；

第 2 行：　为问题添加标签（参阅算法、问题和指标的标签章节）；

第 3 行：　问题的全称；

第 5-9 行：问题的参考文献；

第 12 行：　重定义设定默认属性值的方法；

第 13 行：　设置问题的目标数；

第 14 行：　设置问题的变量数（若未被用户指定）；

第15-16行：设置决策变量的上下界；

第 17 行：　设置决策变量的编码方式；

第 19 行：　重定义计算目标函数的方法；

第 20 行：　计算种群中解的目标值。

除以上代码外，默认的方法 Initialization() 用于随机初始化一个种群，用户可以重定义该方法来指定特殊的种群初始化策略。例如 Sparse_NN.m 将初始化的种群中随机一半的决策变量置零：

```
function Population = Initialization(obj,N)
    if nargin < 2; N = obj.N; end
    PopDec = (rand(N,obj.D)-0.5)*2.*randi([0 1],N,obj.D);
    Population = obj.Evaluation(PopDec);
end
```

默认的方法 CalDec() 将大于上界的决策变量设为上界值、将小于下界的决策变量设为下界值，用户可以重定义该方法来指定特殊的解修复策略。例如 MOKP.m 修复了超过背包容量限制的解，使得该问题无需添加约束函数：

```
function PopDec = CalDec(obj,PopDec)
    C = sum(obj.W,2)/2;
    [~,rank] = sort(max(obj.P./obj.W));
    for i = 1 : size(PopDec,1)
        while any(obj.W*PopDec(i,:)'>C)
            k = find(PopDec(i,rank),1);
            PopDec(i,rank(k)) = 0;
        end
    end
end
```

默认的方法 CalCon()返回零作为解的约束违反值（即解都是满足约束的），用户可以重定义该方法来指定问题的约束。例如 CF4.m 添加了一个约束：

```
function PopCon = CalCon(obj,X)
    t = X(:,2)-sin(6*pi*X(:,1)+2*pi/size(X,2))-0.5*X(:,1)+0.25;
    PopCon = -t./(1+exp(4*abs(t)));
end
```

利用 all(PopCon<=0,2)可确定每个解是否满足所有约束。注意等式约束必须转换为不等式约束来处理，详细方法可参阅该论文的 3.2 节。默认的方法 Evaluation()通过依次调用 CalDec()、CalObj()和 CalCon()来实例化 SOLUTION 对象，同时增加已消耗的评价次数 FE 的值。用户可以重定义该方法在一个函数内完成种群的修复、目标计算和约束计算工作，此时 CalDec()、CalObj()和 CalCon()将不会被调用。例如 MW2.m 同时计算了种群的目标值与约束违反值：

```
function Population = Evaluation(obj,varargin)
    X = varargin{1};
    X=max(min(X,repmat(obj.upper,size(X,1),1)),repmat(obj.lower,size(X,1),1));
    z=1-exp(-10*(X(:,obj.M:end)-(repmat(obj.M:obj.D,size(X,1),1)-1)/obj.D).^2);
    g = 1+sum((1.5+(0.1/obj.D)*z.^2-1.5*cos(2*pi*z)),2);
    PopObj(:,1) = X(:,1);
    PopObj(:,2) = g.*(1-PopObj(:,1)./g);
    L = sqrt(2)*PopObj(:,2)-sqrt(2)*PopObj(:,1);
    PopCon = sum(PopObj,2)-1-0.5*sin(3*pi*l).^8;
    Population = SOLUTION(X,PopObj,PopCon,varargin{2:end});
    obj.FE = obj.FE+length(Population);
end
```

默认的方法 CalGrad()通过有限差分来估计目标函数和约束函数的梯度，用户可以重定义该方法以更准确地计算梯度。用户可以重定义方法 GetOptimum()

来指定问题的最优值，最优值被用于指标值的计算。例如 SOP_F8.m 指定了目标函数的最小值：

```
function R = GetOptimum(obj,N)
    R = -418.9829*obj.D;
end
```

DTLZ2.m 生成了一组前沿面上均匀分布的参考点：

```
function R = GetOptimum(obj,N)
    R = UniformPoint(N,obj.M);
    R = R./repmat(sqrt(sum(R.^2,2)),1,obj.M);
end
```

在不同形状前沿面上的采点方法参阅这里。用户可以重定义方法 GetPF()来指定多目标优化问题的前沿面或可行区域，它们被用于 DrawObj()的可视化中。例如 DTLZ2.m 生成了 2 维和 3 维的前沿面数据：

```
function R = GetPF(obj)
    if obj.M == 2
        R = obj.GetOptimum(100);
    elseif obj.M == 3
        a = linspace(0,pi/2,10)';
        R = {sin(a)*cos(a'),sin(a)*sin(a'),cos(a)*ones(size(a'))};
    else
        R = [];
    end
end
```

MW1.m 生成了可行区域的数据：

```
function R = GetPF(obj)
    [x,y]= meshgrid(linspace(0,1,400),linspace(0,1.5,400));
    z    = nan(size(x));
    fes  = x+y-1-0.5*sin(2*pi*(sqrt(2)*y-sqrt(2)*x)).^8 <= 0;
    z(fes&0.85*x+y>=1) = 0;
    R    = {x,y,z};
end
```

默认的方法 CalMetric()将一个种群与问题的最优值 optimum 传入指标函数中进行计算，用户可以重定义该方法来将不同的变量传入指标函数中。例如 SMMOP1.m 在计算 IGDX 指标时传入问题的最优解集而非前沿面上的参考点：

```
function score = CalMetric(obj,metName,Population)
```

```
    switch metName
        case 'IGDX'
            score = feval(metName,Population,obj.POS);
        otherwise
            score = feval(metName,Population,obj.optimum);
    end
end
```

默认的方法 DrawDec()显示种群的决策向量（用于图形界面中），用户可以重定义该方法来指定特殊的显示方式。例如 TSP.m 显示了种群中最优解的路径：

```
function DrawDec(obj,P)
    [~,best] = min(P.objs);
    Draw(obj.R(P(best).dec([1:end,1]),:),'-k','LineWidth',1.5);
    Draw(obj.R);
end
```

默认的方法 DrawObj()显示种群的目标向量（用于图形界面中），用户可以重定义该方法来指定特殊的显示方式。例如 Sparse_CD.m 添加了坐标轴的标签：

```
function DrawObj(obj,P)
    Draw(P.objs,{'Kernel k-means','Ratio cut',[]});
end
```

其中 Draw()用于显示数据，它位于 PlatEMO\GUI 文件夹中。

## 3. 个体类

一个 SOLUTION 类的对象表示一个个体（即一个解），一组 SOLUTION 类的对象表示一个种群。个体类包含的属性与方法如下：

| 属性 | 赋值方式 | 描述 |
|---|---|---|
| dec | PROBLEM.Evaluation() | 解的决策向量 |
| obj | PROBLEM.Evaluation() | 解的目标值 |
| con | PROBLEM.Evaluation() | 解的约束违反值 |
| add | PROBLEM.Evaluation() | 解的额外属性值（例如速度） |
| 方法 | 描述 | |
| SOLUTION | 生成 SOLUTION 对象数组 | |

| | |
|---|---|
| | 输入一：多个解的决策向量构成的矩阵<br>输入二：多个解的目标值构成的矩阵<br>输入三：多个解的约束违反值构成的矩阵<br>输入四：多个解的额外属性值构成的矩阵<br>输出：SOLUTION 对象数组 |
| decs | 获取多个解的决策向量<br>输入：无<br>输出：多个解的决策向量构成的矩阵 |
| objs | 获取多个解的目标值<br>输入：无<br>输出：多个解的目标值构成的矩阵 |
| cons | 获取多个解的约束违反值<br>输入：无<br>输出：多个解的约束违反值构成的矩阵 |
| adds | 设置并获取多个解的额外属性值<br>输入：默认的额外属性值<br>输出：多个解的额外属性值构成的矩阵 |
| best | 获取种群中可行且最好的解（单目标优化）或可行且非支配的解（多目标优化）<br>输入：无<br>输出：种群中可行且最好的 SOLUTION 对象子数组 |

例如，以下代码产生一个具有十个解的种群，并获取其中最好的解的目标值矩阵：

```
Population = SOLUTION(rand(10,5),rand(10,1),zeros(10,1));
BestObjs   = Population.best.objs
```

注意应只在 PROBLEM 类的方法 Evaluation() 内调用 SOLUTION()。

## 4. 一次完整的运行过程

以下代码利用遗传算法求球面函数的最小值：

```
Alg = GA();
Pro = SOP_F1();
Alg.Solve(Pro);
```

其中代码 Alg.Solve(Pro) 执行时所涉及的函数调用过程如下图所示。




## 5. 指标函数

每个性能指标需要被定义为一个函数并保存在 `PlatEMO\Metrics` 文件夹中。例如 `IGD.m` 的代码为：

```
function score = IGD(Population,optimum)
% <min> <multi/many> <real/integer/label/binary/permutation>
<large/none>       <constrained/none>       <expensive/none>
<multimodal/none> <sparse/none> <dynamic/none> <robust/none>
% Inverted generational distance

%------------------------ Reference -----------------------
% C. A. Coello Coello and N. C. Cortes, Solving
% multiobjective optimization problem using an artificial
% immune system, Genetic Programming and Evolvable
```

```
9  % Machines, 2005, 6(2): 163-190.
10 %--------------------------------------------------------------
11
12     PopObj = Population.best.objs;
13     if size(PopObj,2) ~= size(optimum,2)
14         score = nan;
15     else
16         score = mean(min(pdist2(optimum,PopObj),[],2));
17     end
18 end
```

各行代码的功能如下：

第 1 行：　函数声明，其中第一个输入为一个种群（即一个 SOLUTION 对象数组）、第二个输入为问题的最优值（即问题的 optimum 属性）、输出为种群的指标值；

第 2 行：　为指标添加标签（参阅*算法、问题和指标的标签*章节）；注意标签 <min>或<max>必须为第一个标签；

第 3 行：　指标的全称；

第 5-10 行：指标的参考文献；

第 12 行：　获取种群中最好的解（可行且非支配的解）的目标值矩阵；

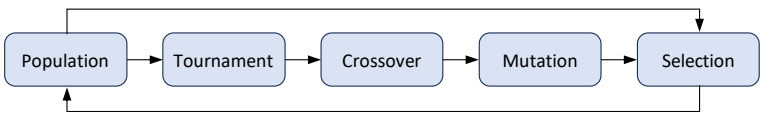
第13-14行：若种群不存在可行解则返回 nan；

第15-16行：否则返回可行且非支配的解的指标值。

## 6. 创建 NeuroEA 算法

NeuroEA 提供了一种创建新算法的灵活框架，它通过有权有向循环图来定义算法。图中每个节点表示一个种群处理模块如交叉、变异和选择，图中每条边决定了解在节点之间的传递方向和比例。每个节点包含许多可以自动训练的参数，一个充分训练的 NeuroEA 算法可以在训练问题上具有突出的性能。

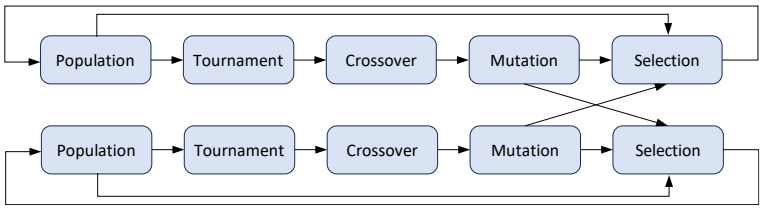
一个 NeuroEA 算法通过一个 BLOCK 对象数组和一个邻接矩阵来表示。例如一个具有如下形式的 NeuroEA 算法




可以使用如下代码创建和运行：

```
addpath('Algorithms\NeuroEA');
Blocks = [Block_Population
          Block_Tournament(200,10)
          Block_Crossover(2,5)
          Block_Mutation(5)
          Block_Selection(100)];
Graph = [0 1 0 0 1
         0 0 1 0 0
         0 0 0 1 0
         0 0 0 0 1
         1 0 0 0 0];
platemo('algorithm',{@NeuroEA,Blocks,Graph},'problem',@SOP_F1);
```

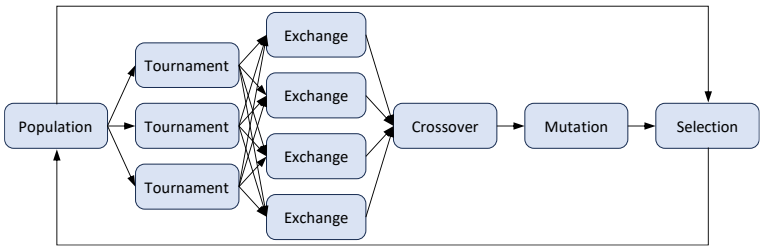
一个具有如下形式的多种群 NeuroEA 算法




可以使用如下代码创建和运行：

```
addpath('Algorithms\NeuroEA');
Blocks = [Block_Population
          Block_Tournament(100,10)
          Block_Crossover(2,5)
          Block_Mutation(5)
          Block_Selection(100)
          Block_Population
          Block_Tournament(100,10)
          Block_Crossover(2,5)
          Block_Mutation(5)
          Block_Selection(100)];
Graph = [0 1 0 0 1 0 0 0 0 0
         0 0 1 0 0 0 0 0 0 0
         0 0 0 1 0 0 0 0 0 0
         0 0 0 0 1 0 0 0 0 1
         1 0 0 0 0 0 0 0 0 0
         0 0 0 0 0 0 1 0 0 1
         0 0 0 0 0 0 0 1 0 0
         0 0 0 0 0 0 0 0 1 0
         0 0 0 0 0 0 0 0 0 1
         0 0 0 0 0 1 0 0 0 0];
platemo('algorithm',{@NeuroEA,Blocks,Graph},'problem',@SOP_F1);
```

一个具有如下形式的复杂 NeuroEA 算法




可以使用如下代码创建和运行：

```
addpath('Algorithms\NeuroEA');
Blocks = [Block_Population
          Block_Tournament(200,10)
          Block_Tournament(200,10)
          Block_Tournament(200,10)
          Block_Exchange(3)
          Block_Exchange(3)
          Block_Exchange(3)
          Block_Exchange(3)
          Block_Crossover(2,5)
          Block_Mutation(5)
          Block_Selection(100)];
Graph = [0 1 1 1 0 0 0 0 0 0 1
         0 0 0 0 1/4 1/4 1/4 1/4 0 0 0
         0 0 0 0 1/4 1/4 1/4 1/4 0 0 0
         0 0 0 0 1/4 1/4 1/4 1/4 0 0 0
         0 0 0 0 0 0 0 0 1 0 0
         0 0 0 0 0 0 0 0 1 0 0
         0 0 0 0 0 0 0 0 1 0 0
         0 0 0 0 0 0 0 0 1 0 0
         0 0 0 0 0 0 0 0 0 1 0
         0 0 0 0 0 0 0 0 0 0 1
         1 0 0 0 0 0 0 0 0 0 0];
platemo('algorithm',{@NeuroEA,Blocks,Graph},'problem',@SOP_F1);
```

NeuroEA 算法中的每个模块是模块类的一个实例。一个模块类需要被定义为 BLOCK 类的子类并保存在 PlatEMO\Algorithms\NeuroEA 文件夹中。模块类包含的属性与方法如下：

| 属性 | 赋值方式 | 描述 |
|---|---|---|
| parameter | ParameterSet() | 模块的参数 |
| lower | 构造函数 | 每个参数的下界 |
| upper | 构造函数 | 每个参数的上界 |

| output | Main() | 模块当前的输出种群 |
|---|---|---|
| nextOut | Gather() | 下个输出的解在种群中的编号 |
| trainTime | 用户 | 已训练次数 |
| 方法 | 是否可重定义 | 描述 |
| 构造函数 | 必须 | 设置由用户指定的属性值<br>输入：超参数的设定值<br>输出：BLOCK 对象 |
| Main | 必须 | 模块的主体部分<br>输入一：PROBLEM 对象<br>输入二：该模块的所有前驱模块<br>输入三：从每个前驱模块获取的解的比例<br>输出：无 |
| ParameterAssign | 可以 | 根据模块的参数确定模块的属性值<br>输入：无<br>输出：无 |
| ParameterSet | 不可 | 设置多个模块的参数<br>输入：多个模块的参数构成的向量<br>输出：无 |
| parameters | 不可 | 获取多个模块的参数<br>输入：无<br>输出：多个模块的参数构成的向量 |
| lowers | 不可 | 获取多个模块的参数的下界<br>输入：无<br>输出：所有参数的下界构成的向量 |
| uppers | 不可 | 获取多个模块的参数的上界<br>输入：无<br>输出：所有参数的上界构成的向量 |
| Gather | 不可 | 从所有前驱模块获取解<br>输入一：PROBLEM 对象<br>输入二：该模块的所有前驱模块<br>输入三：从每个前驱模块获取的解的比例<br>输入四：获取 SOLUTION 对象还是决策变量<br>输入五：值 $k$，获取的解的数目必须是 $k$ 的倍数<br>输出：SOLUTION 对象数组或决策变量矩阵 |
| Validity | 不可 | 检查 NeuroEA 算法的合法性<br>输入：邻接矩阵<br>输出：无 |

每个模块类需要继承 BLOCK 类并重定义构造函数和方法 Main()。例如 Block_Mutation.m 的代码为

```
1  classdef Block_Mutation < BLOCK
2  % Unified mutation for real variables
3  % nSets --- 5 --- Number of parameter sets
4
5      properties
6          nSets;
7          Weight;
8          Fit;
9          nDec = 1;
10     end
11     methods
12         function obj = Block_Mutation(nSets)
13             obj.nSets = nSets;
14             obj.lower = repmat([0 1e-20],1,nSets);
15             obj.upper = repmat([1 5],1,nSets);
16             obj.parameter = unifrnd(obj.lower,ones(1,2*nSets));
17             obj.ParameterAssign();
18         end
19         function ParameterAssign(obj)
20             obj.Weight = reshape(obj.parameter,[],obj.nSets)';
21             obj.Weight(:,end) = obj.Weight(:,end)./obj.nDec;
22             obj.Weight = [obj.Weight;0,max(0,1-sum(obj.Weight(:,end)))];
23             obj.Fit = cumsum(obj.Weight(:,end));
24             obj.Fit = obj.Fit./max(obj.Fit);
25         end
26         function Main(obj,Problem,Precursors,Ratio)
27             ParentDec = obj.Gather(Problem,Precursors,Ratio,2,1);
28             if size(ParentDec,2) ~= obj.nDec
29                 obj.nDec = size(ParentDec,2);
30                 obj.ParameterAssign();
31             end
32             r = ParaSampling(size(ParentDec),obj.Weight(:,1),obj.Fit);
33             obj.output = ParentDec + repmat(Problem.upper-...
                           Problem.lower,size(ParentDec,1),1).*r;
34         end
35     end
36 end
```

各行代码的功能如下：

第 1 行：　继承 BLOCK 类；

第 2 行：　模块的全称；

第 3 行：　超参数名 --- 默认值 --- 超参数描述，将会显示在图形界面的模块

设置面板中；

第5-10行：模块的特有属性；

第12行：重定义构造函数；

第13行：设置一个特有属性的值；

第14-15行：设置模块参数的上下界；

第16行：随机产生模块参数；

第17行：调用方法 `ParameterAssign()` 来根据模块参数确定模块的属性值；

第19-25行：重定义属性值设置的方法；该方法会在方法 `ParameterSet()` 内被自动调用；

第26行：重定义模块主体流程的方法；

第27行：从所有前驱模块获取解的决策变量矩阵；

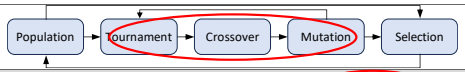
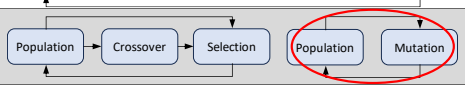
第28-33行：通过变异生成新解的决策变量矩阵；该方法会在 `NeuroEA.m` 内被自动调用。

平台当前提供了七种用于创建 NeuroEA 算法的模块(即 `BLOCK` 类的子类)，包括：

| 模块 | 描述 |
|---|---|
| `Block_Population` | 仅用于存储解，没有任何处理过程 |
| `Block_Crossover` | 基于多个解的交叉产生一个新解 |
| `Block_Mutation` | 变异一个解 |
| `Block_Exchange` | 基于多个解的交换产生一个新解 |
| `Block_Kopt` | 翻转一个解的部分变量的顺序，主要用于序列优化 |
| `Block_Tournament` | 联赛选择，主要用于交配池选择 |
| `Block_Selection` | 保留部分解，主要用于环境选择 |

这些模块及其超参数的详细介绍可参阅该论文。这些模块可以被任意连接以创建 NeuroEA 算法，但需要避免以下非法情形：

| 非法情形 | 示例 |
|---|---|
| 第一个模块不是种群模块 |  |
| 图中不包含任何算子模块 |  |
| 一个模块没有前驱模块 |  |
| 一个模块没有后继模块 |  |
| 一个模块有自环 |  |

| | |
|---|---|
| 一个循环中没有种群模块 |  |
| 图不是强连通图 |  |

通过调用方法 `Validity()` 可检查 NeuroEA 算法的合法性。例如以下代码检查了一个由 `Blocks` 和 `Graph` 定义的 NeuroEA 算法的合法性，且输出由 `Validity()` 抛出的错误 `err` 中存储的非法模块信息：

```
try
    Block.Validity(Graph);
catch err
    switch err.identifier
        case 'BLOCK:NoPopulation'

        case 'BLOCK:NoOperator'

        case 'BLOCK:NoInput'
            str2num(err.cause{1}.message)
        case 'BLOCK:NoOutput'
            str2num(err.cause{1}.message)
        case 'BLOCK:SelfLoop'
            str2num(err.cause{1}.message)
        case 'BLOCK:InfLoop'
            str2num(err.cause{1}.message)
        case 'BLOCK:Isolation'
            str2num(err.cause{1}.message)
    end
end
```

`BLOCK` 类的构造函数的输入是模块的超参数，它们决定了模块的一些特有属性的值。此外，属性 `parameter` 存储了模块的参数，它们也决定了一些特有属性的值且可以被训练以显著提升算法的性能。训练过程可以在创造模块中或使用以下代码实现：

```
addpath('Algorithms\NeuroEA');
Blocks = [Block_Population
          Block_Tournament(200,10)
          Block_Crossover(2,5)
          Block_Mutation(5)
          Block_Selection(100)];
Graph = [0 1 0 0 1
         0 0 1 0 0
```

```
          0 0 0 1 0
          0 0 0 0 1
          1 0 0 0 0];
function y = Fcn(x,data)
    data{1}.ParameterSet(x);
    for i = 1 : 3
        [~,obj] = platemo('algorithm',...
                          {@NeuroEA,data{1},data{2}},...
                          'problem',@SOP_F1,...
                          'outputFcn',@(~,~)[]);
        s(i) = min(obj);
    end
    y = mean(s);
end
platemo('algorithm',@GA,'objFcn',@Fcn,...
         'lower',Blocks.lowers,...
         'upper',Blocks.uppers,...
         'D',length(Blocks.lowers),...
         'data',{Blocks,Graph},'N',30,'save',1);
```

以上代码将 NeuroEA 算法的训练视为一个优化问题并利用`@GA` 求解，其中决策变量是所有模块的参数，目标函数 `Fcn()`定义为该 NeuroEA 算法在`@SOP_F1`上三次优化性能的平均值。

# 五　算法列表

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | ABC | Artificial bee colony algorithm | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 2 | AB-SAEA | Adaptive Bayesian based surrogate-assisted evolutionary algorithm | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 3 | AC-MMEA | Adaptive merging and coordinated offspring generation based multi-modal multi-objective evolutionary algorithm | | √ | | √ | √ | | | | √ | | | √ | √ | | | | |
| 4 | ACO | Ant colony optimization | √ | | | | | | | √ | √ | | | | | | | | |
| 5 | Adam | Adaptive moment estimation | √ | | | √ | | | | | √ | | | | | | | | |
| 6 | AdaW | Evolutionary algorithm with adaptive weights | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 7 | ADSAPSO | Adaptive dropout based surrogate-assisted particle swarm optimization | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 8 | AE-NSGA-II | Autoencoding NSGA-II | | √ | | √ | √ | √ | √ | √ | | √ | | | | √ | | | |
| 9 | AESSPSO | Adaptive exploration state-space particle swarm optimization | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 10 | AFSEA | Adjoint feature-selection-based evolutionary algorithm | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 11 | AGE-II | Approximation-guided evolutionary multi-objective algorithm II | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 12 | AGE-MOEA | Adaptive geometry estimation-based many-objective evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 13 | AGE-MOEA-II | Adaptive geometry estimation-based many-objective evolutionary algorithm II | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 14 | AGSEA | Automated guiding vector selection-based evolutionary algorithm | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 15 | AMG-PSL | Adaptive multi-granular Pareto-optimal subspace learning | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 16 | A-NSGA-III | Adaptive NSGA-III | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 17 | APSEA | Adaptive population sizing based evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 18 | AR-MOEA | Adaptive reference points based multi-objective evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 19 | AutoV | Automated design of variation operators | √ | √ | | √ | √ | | | | √ | √ | | | | | | | |
| 20 | AVG-SAEA | Adaptive variable grouping based surrogate-assisted evolutionary algorithm | | √ | | √ | √ | | | | √ | | √ | | | | | | |
| 21 | BCE-IBEA | Bi-criterion evolution based IBEA | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 22 | BCE-MOEA/D | Bi-criterion evolution based MOEA/D | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 23 | BFGS | A quasi-Newton method proposed by Broyden, Fletcher, Goldfarb, and Shanno | √ | | | √ | | | | | √ | | | | | | | | |
| 24 | BiCo | Bidirectional coevolution constrained multiobjective evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 25 | BiGE | Bi-goal evolution | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 26 | BLEAQII | Bilevel evolutionary algorithm based on quadratic approximations II | | √ | | √ | | | | | | √ | | | | | | √ | |
| 27 | BL-SAEA | Bi-level surrogate modelling based evolutionary algorithm | | √ | | √ | | | | | | √ | | | | | | √ | |
| 28 | BSPGA | Binary space partition tree based genetic algorithm | √ | | | | | | √ | | √ | √ | | | | | | | |
| 29 | C3M | Constraint, multiobjective, multi-stage, multi-constraint evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 30 | CAEAD | Dual-population evolutionary algorithm based on alternative evolution and degeneration | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 31 | CA-MOEA | Clustering based adaptive multi-objective evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 32 | CCGDE3 | Cooperative coevolution GDE3 | | √ | | √ | √ | | | | √ | | | | | | | | |
| 33 | CCMO | Coevolutionary constrained multi-objective optimization framework | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 34 | c-DPEA | Constrained dual-population evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 35 | CGLP | Correlation-guided layered prediction | | √ | | √ | √ | √ | √ | √ | | | | | | √ | | | |
| 36 | CI-EMO | Composite indicator-guided infilling sampling for expensive multi-objective optimization | | √ | √ | √ | | | | | | | √ | | | | | | |
| 37 | CKAnD | Constrained Kriging-assisted AnD | | √ | √ | √ | | | | | | √ | √ | | | | | | |
| 38 | CLIA | Evolutionary algorithm with cascade clustering and reference point incremental learning | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 39 | CMaDPPs | Constrained many-objective optimization with determinantal point processes | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 40 | CMA-ES | Covariance matrix adaptation evolution strategy | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 41 | CMDEIPCM | Constrained multiobjective differential evolution algorithm with an infeasible proportion control mechanism | | √ | | √ | √ | | | | √ | √ | | | | | | | |
| 42 | CMEA | Cooperative multi-population evolutionary algorithm | | √ | | √ | √ | | | | √ | | | √ | √ | | | | |
| 43 | CMEGL | Constrained evolutionary multitasking with global and local auxiliary tasks | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 44 | CMME | Constrained many-objective evolutionary algorithm with enhanced mating and environmental selections | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 45 | CMMO | Coevolutionary multi-modal multi-objective optimization framework | | √ | | √ | √ | √ | √ | √ | | | | √ | | | | | |
| 46 | CMOBR | Constrained multiobjective optimization via both constraint and objective relaxations | | √ | √ | √ | √ | | | | | √ | | | | | | | |
| 47 | CMOCEA-DDAP | Constrained multi-objective co-evolutionary algorithm with a dynamic diversity-enhanced auxiliary population | | √ | | √ | √ | | | | | √ | | | | | | | |
| 48 | CMOCSO | Competitive and cooperative swarm optimization constrained multi-objective optimization algorithm | | √ | | √ | | | | | √ | √ | | | | | | | |
| 49 | CMODE-FTR | Constrained multiobjective differential evolution based on the fusion of two rankings | | √ | | √ | √ | | | | | √ | | | | | | | |
| 50 | CMODRL | Constrained multiobjective optimization via deep reinforcement learning | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 51 | CMOEA-2S | Constrained MOEA with two types of evolution stages | | √ | | √ | | | | | | √ | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 52 | CMOEA-ACRT | Constrained MOEA with comprehensive ranking technique | | √ | | √ | | | | | | √ | | | | | | | |
| 53 | CMOEA-AOP | Automated operator portfolio based constrained MOEA | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 54 | CMOEA-CD | Constraint-Pareto dominance and diversity enhancement strategy based constrained MOEA | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 55 | C-MOEA/D | Constraint-MOEA/D | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 56 | CMOEA-MS | Constrained multiobjective evolutionary algorithm with multiple stages | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 57 | CMOEA-MSG | Multi-stage constrained multi-objective evolutionary algorithm | | √ | | √ | √ | | | | | √ | | | | | | | |
| 58 | CMOEA-RDA | Constrained MOEA with role dynamic adjustment | | √ | | √ | | | | | | √ | | | | | | | |
| 59 | CMOEBOD | Constrained multiobjective evolutionary Bayesian optimization based on decomposition | | √ | √ | √ | | | | | | √ | √ | | | | | | |
| 60 | CMOEMT | Constrained multi-objective optimization based on evolutionary multitasking optimization | | √ | | √ | | | | | | √ | | | | | | | |
| 61 | CMOES | Constrained multi-objective optimization based on even search | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 62 | CMOFLCS | Constrained multi-objective optimization with fuzzy logic control system | | √ | | √ | | | | | | √ | | | | | | | |
| 63 | CMOPSO | Competitive mechanism based multi-objective particle swarm optimizer | | √ | | √ | √ | | | | | | | | | | | | |
| 64 | CMOQLMT | Constrained multi-objective optimization based on Q-learning and multitasking | | √ | | √ | | | | | | √ | | | | | | | |
| 65 | CMOSMA | Constrained multi-objective evolutionary algorithm with self-organizing map | | √ | √ | √ | √ | | | | | √ | | | | | | | |
| 66 | CNSDE/DVC | Constrained nondominated sorting differential evolution based on decision variable classification | | √ | | √ | √ | | | | | | | | | | | | √ |
| 67 | CoMMEA | Coevolutionary multimodal multi-objective evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | | | √ | | | | | |
| 68 | CPCMO | Constraint, multiobjective, multi-stage, multi-constraint evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 69 | CPS-MOEA | Classification and Pareto domination based multi-objective evolutionary | | √ | | √ | √ | | | | | | √ | | | | | | |
| 70 | CSEA | Classification based surrogate-assisted evolutionary algorithm | | √ | √ | √ | | | | | | | √ | | | | | | |
| 71 | CSEMT | Constraints separation based evolutionary multitasking | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 72 | CSO | Competitive swarm optimizer | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 73 | C-TAEA | Two-archive evolutionary algorithm for constrained MOPs | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 74 | C-TSEA | Constrained two-stage evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 75 | DAEA | Duplication analysis based evolutionary algorithm | | √ | | | | | √ | | | | | | | | | | |
| 76 | DBEMTO | Double-balanced evolutionary multi-task optimization | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 77 | DCNSGA-III | Dynamic constrained NSGA-III | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 78 | DE | Differential evolution | √ | | | √ | √ | | | | √ | √ | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 79 | DEA-GNG | Decomposition based evolutionary algorithm guided by growing neural gas | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 80 | DGEA | Direction guided evolutionary algorithm | | √ | √ | √ | √ | | | | √ | | | | | | | | |
| 81 | DirHV-EI | Expected direction-based hypervolume improvement | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 82 | DISK | Distribution-based Kriging-assisted evolutionary algorithm | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 83 | DISKplus | Distribution-based Kriging-assisted constrained evolutionary algorithm | | √ | √ | √ | √ | | | | | √ | √ | | | | | | |
| 84 | DKCA | Dynamic knowledge-guided coevolutionary algorithm | | √ | | √ | | | √ | | √ | √ | | | √ | | | | |
| 85 | DM-MOEA | Dual model based multi-objective evolutionary algorithm | | √ | | √ | √ | | √ | | √ | √ | | | √ | √ | | | |
| 86 | DMOEA-eC | Decomposition-based multi-objective evolutionary algorithm with the e-constraint framework | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 87 | dMOPSO | MOPSO based on decomposition | | √ | | √ | √ | | | | | | | | | | | | |
| 88 | DN-NSGA-II | Decision space based niching NSGA-II | | √ | | √ | √ | | | | | | | √ | | | | | |
| 89 | DNSGA-II | Dynamic NSGA-II | | √ | | √ | √ | √ | √ | √ | | | | | | √ | | | |
| 90 | DOA | Dandelion optimization algorithm | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 91 | DPCPRA | Dual-population with dynamic constraint processing and resource allocating | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 92 | DP-PPS | Tri-population based push and pull search | | √ | | √ | | | | | | √ | | | | | | | |
| 93 | DPVAPS | Dual-population with variable auxiliary population size | | √ | | √ | √ | | | | √ | √ | | | | | | | |
| 94 | DRLOS-EMCMO | EMCMO with deep reinforcement learning-assisted operator selection | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 95 | DRL-SAEA | Deep reinforcement learning-based expensive constrained evolutionary algorithm | | √ | | √ | | | | | | √ | √ | | | | | | |
| 96 | DSOCOL | Dual-swarm optimizer with collaborative orthogonal learning | | √ | | √ | | | | | √ | √ | | | | | | | |
| 97 | DSPCMDE | Dynamic selection preference-assisted constrained multiobjective differential evolution | | √ | | √ | √ | | | | | √ | | | | | | | |
| 98 | DSSEA | Dynamic subspace search-based evolutionary algorithm | | √ | √ | √ | √ | | | | √ | √ | | | | | | | |
| 99 | DVCEA | Decision variables classification-based evolutionary algorithm | | √ | √ | √ | √ | | | | √ | √ | | | | | | | |
| 100 | DWU | Dominance-weighted uniformity multi-objective evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 101 | EAG-MOEA/D | External archive guided MOEA/D | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 102 | ECPO | Electric charged particles optimization | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 103 | EDN-ARMOEA | Efficient dropout neural network based AR-MOEA | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 104 | EFR-RR | Ensemble fitness ranking with a ranking restriction scheme | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 105 | EGES | Efficient grouping evolutionary search | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 106 | EGO | Efficient global optimization | √ | | | √ | √ | | | | | | √ | | | | | | |
| 107 | EIM-EGO | Expected improvement matrix based efficient global optimization | | √ | | √ | √ | | | | | | √ | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 108 | EMCMMS | Evolutionary multitasking with a cooperative multistep mutation strategy | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 109 | EMCMO | Evolutionary multitasking-based constrained multiobjective optimization | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 110 | EMMOEA | Expensive multi-/many-objective evolutionary algorithm | | √ | | √ | √ | | | | | | √ | | | | | | |
| 111 | e-MOEA | Epsilon multi-objective evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 112 | EMOSKT | Evolutionary multi-objective optimization with sparsity knowledge transfer | | √ | | √ | | | √ | | √ | √ | | | √ | | √ | | |
| 113 | EM-SAEA | Ensemble-based surrogate model-assisted evolutionary algorithm | | √ | √ | √ | | | | | | √ | √ | | | | | | |
| 114 | EMyO/C | Evolutionary many-objective optimization algorithm with clustering-based | | √ | √ | √ | √ | | | | | | | | | | | | |
| 115 | ENS-MOEA/D | Ensemble of different neighborhood sizes based MOEA/D | | √ | √ | √ | √ | | | | | | | | | | | | |
| 116 | ESBCEO | Bayesian co-evolutionary optimization based entropy search | | √ | | √ | | | | | | | √ | | | | | | |
| 117 | FDSEA | Frequency domain search-based evolutionary algorithm | | √ | | √ | √ | | | | √ | | | | | | | | |
| 118 | FDV | Fuzzy decision variable framework with various internal optimizers | | √ | √ | √ | √ | | | | √ | | | | | | | | |
| 119 | FEP | Fast evolutionary programming | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 120 | FLEA | Fast sampling based evolutionary algorithm | | √ | √ | √ | | | | | √ | | | | | | | | |
| 121 | FRCG | Fletcher-Reeves conjugate gradient | √ | | | √ | | | | | √ | | | | | | | | |
| 122 | FRCGM | Fletcher-Reeves conjugate gradient (for multi-objective optimization) | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 123 | FROFI | Feasibility rule with the incorporation of objective function information | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 124 | GA | Genetic algorithm | √ | | | √ | √ | √ | √ | √ | √ | √ | | | | | | | |
| 125 | GCNMOEA | Graph convolutional network based multi-objective evolutionary algorithm | | √ | | √ | √ | | | | | | | | | | | | |
| 126 | GDE3 | Generalized differential evolution 3 | | √ | | √ | √ | | | | | √ | | | | | | | |
| 127 | GDVTSF | Generational difference vector based tri-entropy structure framework | | √ | | √ | | | | | √ | | | | | | | | |
| 128 | GFM-MOEA | Generic front modeling based multi-objective evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 129 | GLMO | Grouped and linked mutation operator algorithm | | √ | | √ | √ | | | | √ | | | | | | | | |
| 130 | g-NSGA-II | g-dominance based NSGA-II | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 131 | GPSO | Gradient based particle swarm optimization algorithm | √ | | | √ | | | | | √ | √ | | | | | | | |
| 132 | GPSOM | Gradient based particle swarm optimization algorithm (for multi-objective optimization) | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 133 | GrEA | Grid-based evolutionary algorithm | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 134 | GWASF-GA | Global weighting achievement scalarizing function genetic algorithm | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 135 | GWO | Grey wolf optimizer | √ | | | √ | √ | | | | √ | √ | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| 136 | HEA | Hyper-dominance based evolutionary algorithm | | √ | √ | √ | | | √ | √ | | | | | | | | | |
| 137 | HeE-MOEA | Multiobjective evolutionary algorithm with heterogeneous ensemble based infill criterion | | √ | | √ | √ | | | | | | √ | | | | | | |
| 138 | HHC-MMEA | Hybrid hierarchical clustering based multi-modal multi-objective evolutionary algorithm | | √ | | √ | | | | | √ | | | √ | √ | | | | |
| 139 | hpaEA | Hyperplane assisted evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 140 | HREA | Hierarchy ranking based evolutionary algorithm | | √ | | √ | √ | | | | | | | √ | | | | | |
| 141 | HypE | Hypervolume estimation algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 142 | IBEA | Indicator-based evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 143 | ICMA | Indicator based constrained multi-objective algorithm | | √ | | √ | √ | | | | | √ | | | | | | | |
| 144 | I-DBEA | Improved decomposition-based evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 145 | ILCMO | Indicator-based evolutionary algorithm for large-scale constrained multi-objective optimization | | √ | √ | √ | √ | | | | √ | √ | | | | | | | |
| 146 | IM-C-MOEA/D | Inverse modeling constrained MOEA/D | | √ | | √ | √ | | | | √ | √ | | | | | | | |
| 147 | IM-MOEA | Inverse modeling based multiobjective evolutionary algorithm | | √ | | √ | √ | | | | √ | | | | | | | | |
| 148 | IM-MOEA/D | Inverse modeling MOEA/D | | √ | | √ | √ | | | | √ | | | | | | | | |
| 149 | IMODE | Improved multi-operator differential evolution | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 150 | IMTCMO | Improved evolutionary multitasking-based CMOEA | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 151 | IMTCMO_BS | Improved evolutionary multitasking-based CMOEA with bidirectional sampling | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 152 | I-SIBEA | Interactive simple indicator-based evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 153 | Izui | An aggregative gradient based multi-objective optimizer proposed by Izui et al. | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 154 | KAnD | Kriging-assisted AnD | | √ | √ | √ | | | | | | | √ | | | | | | |
| 155 | KLEA | Knowledge learning-based evolutionary algorithm | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 156 | KL-NSGA-II | Knowledge learning based NSGA-II | | √ | | √ | √ | √ | √ | √ | | √ | | | | √ | | | |
| 157 | KMA | Komodo mlipir algorithm | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 158 | KnEA | Knee point driven evolutionary algorithm | | | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 159 | K-RVEA | Surrogate-assisted RVEA | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 160 | KTA2 | Kriging-assisted Two_Arch2 | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 161 | KTS | Kriging-assisted evolutionary algorithm with two search modes | | √ | √ | | √ | | | | | √ | √ | | | | | | |
| 162 | L2SMEA | Linear subspace surrogate modeling assisted evolutionary algorithm | √ | | | √ | | | | | | | √ | | | | | | |
| 163 | LCMEA | Large-scale constrained multi-objective evolutionary algorithm | | √ | | √ | | | | | √ | √ | | | | | | | |
| 164 | LCSA | Linear combination-based search algorithm | | √ | √ | √ | √ | | | | √ | | | | | | | | |
| 165 | LDS-AF | Low-dimensional surrogate aggregation function | | √ | | √ | √ | | | | √ | | √ | | | | | | |
| 166 | LERD | Large-scale evolutionary algorithm with reformulated decision variable analysis | | √ | √ | √ | | | | | √ | | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| 167 | LMEA | Evolutionary algorithm for large-scale many-objective optimization | | √ | √ | √ | √ | | | | √ | | | | | | | | |
| 168 | LMOCSO | Large-scale multi-objective competitive swarm optimization algorithm | | √ | √ | √ | √ | | | | √ | √ | | | | | | | |
| 169 | LMOEA-DS | Large-scale evolutionary multi-objective optimization assisted by directed sampling | | √ | | √ | √ | | | | √ | | | | | | | | |
| 170 | LMPFE | Evolutionary algorithm with local model based Pareto front estimation | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 171 | LRMOEA | Large-scale robust multi-objective evolutionary algorithm | | √ | | √ | | | √ | | √ | √ | | | √ | | | | √ |
| 172 | LSMOF | Large-scale multi-objective optimization framework with NSGA-II | | √ | | √ | √ | | | | √ | | | | | | | | |
| 173 | MaOEA-CSS | Many-objective evolutionary algorithms based on coordinated selection | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 174 | MaOEA-DDFC | Many-objective evolutionary algorithm based on directional diversity and favorable convergence | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 175 | MaOEA-HAP | Hyper-curvature balanced indicator and adaptive phase exploration co-driven many-objective evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 176 | MaOEA/IGD | IGD based many-objective evolutionary algorithm | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 177 | MaOEA/IT | Many-objective evolutionary algorithms based on an independent two-stage | | √ | √ | √ | √ | | | | | √ | | | | | | | |
| 178 | MaOEA-R&D | Many-objective evolutionary algorithm based on objective space reduction | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 179 | MaOTSPP | MaOEA based on two-stage probabilistic penalty | | √ | √ | √ | | | | | | √ | √ | | | | | | |
| 180 | MASSR-CMODRL | Constrained multi-objective optimization with MASSR states | | √ | | √ | | | | | | √ | | | | | | | |
| 181 | MASSR-MADE | Multi-agent differential evolution with MASSR states | √ | | | √ | | | | | | | | | | | | | |
| 182 | MASSR-MAMOEA/D | Multi-agent self-supervised state representation with MOEA/D | | √ | | √ | | | | | | | | | | | | | |
| 183 | MASSR-MARL-AC-S | Sparse multi-objective optimization with MASSR states | | √ | | √ | | | √ | | √ | | | | √ | | | | |
| 184 | MCCMO | Multi-population coevolutionary constrained multi-objective optimization | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 185 | MCEA/D | Multiple classifiers-assisted evolutionary algorithm based on decomposition | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 186 | MFEA | Multifactorial evolutionary algorithm | √ | | | √ | √ | √ | √ | √ | √ | | | | | | √ | | |
| 187 | MFEA-II | Multifactorial evolutionary algorithm II | √ | | | √ | √ | √ | √ | √ | √ | | | | | | √ | | |
| 188 | MFFS | Multiform feature selection | | √ | | | | | √ | | | | | | | | | | |
| 189 | MFO-SPEA2 | Multiform optimization framework based on SPEA2 | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 190 | MGCEA | Multi-granularity clustering based evolutionary algorithm | | √ | | √ | | | √ | | √ | √ | | | √ | | | | |
| 191 | MGO | Mountain gazelle optimizer | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 192 | MGSAEA | Multigranularity surrogate-assisted constrained evolutionary algorithm | | √ | | √ | | | | | | √ | √ | | | | | | |
| 193 | MiSACO | Multi surrogate-assisted ant colony optimization | √ | | | √ | √ | | | | | | √ | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| 194 | MMEA-ARM | Adaptive resource management based MMEA | | √ | | √ | √ | | √ | | √ | | | √ | √ | | | | |
| 195 | MMEAPSL | Multimodal multi-objective evolutionary algorithm assisted by Pareto set learning | | √ | | √ | √ | √ | √ | √ | | | | √ | | | | | |
| 196 | MMEA-WI | Weighted indicator-based evolutionary algorithm for multimodal multi-objective optimization | | √ | | √ | √ | | | | | | | √ | | | | | |
| 197 | MMOEA-BH | Multimodal MOEA with block optimization and hybrid clustering | | √ | | √ | √ | √ | √ | √ | | | | √ | | | | | |
| 198 | MMOPSO | MOPSO with multiple search strategies | | √ | | √ | √ | | | | | | | | | | | | |
| 199 | MO_Ring_PSO_SCD | Multiobjective PSO using ring topology and special crowding distance | | √ | | √ | √ | | | | | | | √ | | | | | |
| 200 | MOBCA | Multi-objective besiege and conquer algorithm | | √ | | √ | √ | | | | | | | | | | | | |
| 201 | MOCell | Cellular genetic algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 202 | MOCGDE | Multi-objective conjugate gradient and differential evolution algorithm | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 203 | MO-CMA | Multi-objective covariance matrix adaptation evolution strategy | | √ | | √ | √ | | | | | | | | | | | | |
| 204 | MOEA/CKF | Multi-objective evolutionary algorithm based on cross-scale knowledge fusion | | √ | | √ | | | √ | | √ | √ | | | √ | | | | |
| 205 | MOEA/D | Multiobjective evolutionary algorithm based on decomposition | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 206 | MOEA/D-2WA | MOEA/D with two-type weight vector adjustments | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 207 | MOEA/D-AWA | MOEA/D with adaptive weight adjustment | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 208 | MOEA/D-CMA | MOEA/D with covariance matrix adaptation evolution strategy | | √ | √ | √ | √ | | | | | | | | | | | | |
| 209 | MOEA/D-CMT | MOEA/D with competitive multitasking | | √ | | √ | | | | | | √ | | | | | | | |
| 210 | MOEA/DD | Many-objective evolutionary algorithm based on dominance and decomposition | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 211 | MOEA/D-DAE | MOEA/D with detect-and-escape strategy | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 212 | MOEA/D-DCWV | MOEA/D with distribution control of weight vector set | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 213 | MOEA/D-DE | MOEA/D based on differential evolution | | √ | √ | √ | √ | | | | | | | | | | | | |
| 214 | MOEA/D-DQN | MOEA/D based on deep Q-network | | √ | √ | √ | √ | | | | | | | | | | | | |
| 215 | MOEA/D-DRA | MOEA/D with dynamical resource allocation | | √ | √ | √ | √ | | | | | | | | | | | | |
| 216 | MOEA/D-DU | MOEA/D with a distance based updating strategy | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 217 | MOEA/D-DYTS | MOEA/D with dynamic Thompson sampling | | √ | √ | √ | √ | | | | | | | | | | | | |
| 218 | MOEA/D-EGO | MOEA/D with efficient global optimization | | √ | | √ | √ | | | | | | √ | | | | | | |
| 219 | MOEA/D-EGO-CEI | MOEA/D-EGO with CEI | | √ | √ | √ | | | | | | √ | √ | | | | | | |
| 220 | MOEA/D-FRRMAB | MOEA/D with fitness-rate-rank-based multiarmed bandit | | √ | √ | √ | √ | | | | | | | | | | | | |
| 221 | MOEA/D-M2M | MOEA/D based on MOP to MOP | | √ | | √ | √ | | | | | | | | | | | | |
| 222 | MOEA/D-MRDL | MOEA/D with maximum relative diversity loss | | √ | | √ | √ | | | | | | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| 223 | MOEA/D-PaS | MOEA/D with Pareto adaptive scalarizing approximation | | √ | √ | √ | √ | | | | | | | | | | | | |
| 224 | MOEA/D-PFE | MOEA/D with Pareto front estimation | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 225 | MOEA/D-STM | MOEA/D with stable matching | | √ | √ | √ | √ | | | | | | | | | | | | |
| 226 | MOEA/D-UR | MOEA/D with update when required | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 227 | MOEA/D-URAW | MOEA/D with uniform randomly adaptive weights | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 228 | MOEA/DVA | Multi-objective evolutionary algorithm based on decision variable | | √ | | √ | √ | | | | √ | | | | | | | | |
| 229 | MOEA/D-VOV | MOEA/D with virtual objective vectors | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 230 | MOEA-IB | MOEA with information bottleneck | | √ | √ | √ | | | | | √ | | | | | | | | |
| 231 | MOEA/IGD-NS | Multi-objective evolutionary algorithm based on an enhanced IGD | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 232 | MOEA-NZD | Multi-objective evolutionary algorithm with nonzero detection | | √ | √ | √ | | | | | √ | √ | | | √ | | | | |
| 233 | MOEA-PC | Multiobjective evolutionary algorithm based on polar coordinates | | √ | | √ | √ | | | | | | | | | | | | |
| 234 | MOEA/PSL | Multi-objective evolutionary algorithm based on Pareto optimal subspace | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 235 | MOEA-RE | Multi-objective evolutionary algorithm with robustness enhancement | | √ | | √ | √ | √ | √ | √ | | | | | | | | | √ |
| 236 | MOEA-SNN | Multi-objective evolutionary algorithm based on spiking neural networks | | √ | | √ | | | √ | | √ | | | | √ | | | | |
| 237 | MO-EGS | Multi-objective evolutionary gradient search | | √ | | √ | | | | | √ | | | | | | | | |
| 238 | MO-L2SMEA | Multi-objective linear subspace surrogate modeling assisted evolutionary algorithm | | √ | | √ | | | | | √ | | √ | | | | | | |
| 239 | MOMBI-II | Many objective metaheuristic based on the R2 indicator II | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 240 | MO-MFEA | Multi-objective multifactorial evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | √ | | |
| 241 | MO-MFEA-II | Multi-objective multifactorial evolutionary algorithm II | | √ | | √ | √ | √ | √ | √ | | √ | | | | | √ | | |
| 242 | MOMFEA-SADE | Multi-objective multifactorial evolutionary algorithm with subspace alignment and adaptive differential evolution | | √ | | √ | √ | √ | √ | √ | | √ | | | | | √ | | |
| 243 | MONAS | Multi-objective neural architecture search | | √ | | √ | √ | √ | √ | √ | | | | √ | | | | | |
| 244 | MOPSO | Multi-objective particle swarm optimization | | √ | | √ | √ | | | | | | | | | | | | |
| 245 | MOPSO-CD | MOPSO with crowding distance | | √ | | √ | √ | | | | | | | | | | | | |
| 246 | MOSD | Multiobjective steepest descent | | √ | | √ | | | | | √ | √ | | | | | | | |
| 247 | M-PAES | Memetic algorithm with Pareto archived evolution strategy | | √ | | √ | √ | | | | | | | | | | | | |
| 248 | MP-MMEA | Multi-population multi-modal multi-objective evolutionary algorithm | | √ | | √ | √ | | | | √ | | | √ | √ | | | | |
| 249 | MPSO/D | Multi-objective particle swarm optimization algorithm based on decomposition | | √ | √ | √ | √ | | | | | | | | | | | | |
| 250 | MSCEA | Multi-stage constrained multi-objective evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| 251 | MSCMO | Multi-stage constrained multi-objective evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 252 | MSEA | Multi-stage multi-objective evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 253 | MSKEA | Multi-stage knowledge-guided evolutionary algorithm | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 254 | MSOPS-II | Multiple single objective Pareto sampling II | | √ | √ | √ | √ | | | | | √ | | | | | | | |
| 255 | MTCMO | Multitasking constrained multi-objective optimization | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 256 | MTDE-MKTA | Multitasking differential evolution with multiple knowledge types and transfer adaptation | | √ | | √ | √ | √ | √ | √ | | √ | | | | | √ | | |
| 257 | MTEA/D-DN | Multiobjective multitask evolutionary algorithm based on decomposition with dual neighborhoods | | √ | | √ | √ | √ | √ | √ | | √ | | | | | √ | | |
| 258 | MTS | Multiple trajectory search | | √ | | √ | √ | | | | | | | | | | | | |
| 259 | MultiObjective EGO | Multi-objective efficient global optimization | | √ | | √ | √ | | | | | √ | √ | | | | | | |
| 260 | MVPA | Most valuable player algorithm | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 261 | MyO-DEMR | Many-objective differential evolution with mutation restriction | | √ | √ | √ | √ | | | | | | | | | | | | |
| 262 | NBLEA | Nested bilevel evolutionary algorithm | | √ | | √ | | | | | | √ | | | | | | √ | |
| 263 | NelderMead | The Nelder-Mead algorithm | √ | | | √ | | | | | | | | | | | | | |
| 264 | NMPSO | Novel multi-objective particle swarm optimization | | √ | √ | √ | √ | | | | | | | | | | | | |
| 265 | NNDREA-MO | Evolutionary algorithm with neural network-based dimensionality reduction (multi-objective) | | √ | | | | | √ | | √ | √ | | | √ | | | | |
| 266 | NNDREA-SO | Evolutionary algorithm with neural network-based dimensionality reduction (single-objective) | √ | | | | | | √ | | √ | √ | | | √ | | | | |
| 267 | NNIA | Nondominated neighbor immune algorithm | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 268 | NRV-MOEA | Adaptive normal reference vector-based multi- and many-objective evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 269 | NSBiDiCo | Non-dominated sorting bidirectional differential coevolution algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 270 | NSGA-II | Nondominated sorting genetic algorithm II | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 271 | NSGA-II+ARSBX | NSGA-II with adaptive rotation based simulated binary crossover | | √ | | √ | √ | | | | | √ | | | | | | | |
| 272 | NSGA-II-conflict | NSGA-II with conflict-based partitioning strategy | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 273 | NSGA-II-DTI | NSGA-II of Deb's type I robust version | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | √ |
| 274 | NSGA-III | Nondominated sorting genetic algorithm III | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 275 | NSGAIII-EHVI | NSGA-III with expected hypervolume improvement | | √ | √ | √ | | | | | | | √ | | | | | | |
| 276 | NSGA-II/SDR | NSGA-II with strengthened dominance relation | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 277 | NSLS | Multiobjective optimization framework based on nondominated sorting and local search | | √ | | √ | √ | | | | | | | | | | | | |
| 278 | NUCEA | Non-uniform clustering based evolutionary algorithm | | √ | | √ | | | √ | | √ | √ | | | √ | | | | |
| 279 | OFA | Optimal foraging algorithm | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 280 | one-by-one EA | Many-objective evolutionary algorithm using a one-by-one selection | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 281 | OSP-NSDE | Non-dominated sorting differential evolution with prediction in the objective space | | √ | | √ | √ | | | | | | | | | | | | |
| 282 | ParEGO | Efficient global optimization for Pareto optimization | | √ | | √ | √ | | | | | | √ | | | | | | |
| 283 | PB-NSGA-III | NSGA-III based on Pareto based bi-indicator infill sampling criterion | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 284 | PB-RVEA | RVEA based on Pareto based bi-indicator infill sampling criterion | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 285 | PC-SAEA | Pairwise comparison based surrogate-assisted evolutionary algorithm | | √ | √ | | | | | | | | √ | | | | | | |
| 286 | PEA | Pareto-based Kriging-assisted constrained multiobjective evolutionary algorithm | | √ | | √ | √ | | | | | √ | √ | | | | | | |
| 287 | PEAplus | Pareto-based Kriging-assisted constrained multiobjective evolutionary algorithm plus | | √ | | √ | √ | | | | | √ | √ | | | | | | |
| 288 | PeEA | Pareto front shape estimation based evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 289 | PESA-II | Pareto envelope-based selection algorithm II | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 290 | PICEA-g | Preference-inspired coevolutionary algorithm with goals | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 291 | PIEA | Performance indicator-based evolutionary algorithm | | √ | √ | √ | | | | | | | √ | | | | | | |
| 292 | PIMD | Probability and mapping crowding distance | | √ | √ | √ | | | | | | | √ | | | | | | |
| 293 | PM-MOEA | Pattern mining based multi-objective evolutionary algorithm | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 294 | POCEA | Paired offspring generation based constrained evolutionary algorithm | | √ | | √ | √ | | | | √ | √ | | | | | | | |
| 295 | PPS | Push and pull search algorithm | | √ | √ | √ | √ | | | | | √ | | | | | | | |
| 296 | PRCEA | Promising region-guided large-scale constrained MOEA | | √ | √ | √ | √ | | | | √ | √ | | | | | | | |
| 297 | PRDH | Problem reformulation and duplication handling | | √ | | | | | √ | | | | | | | | | | |
| 298 | PREA | Promising-region based EMO algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 299 | PSO | Particle swarm optimization | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 300 | REMO | Expensive multiobjective optimization by relation learning and prediction | | √ | √ | √ | | | | | | | √ | | | | | | |
| 301 | RGA-M1-2 | Real-coded genetic algorithm with framework M1-2 | | √ | | √ | | | | | | √ | √ | | | | | | |
| 302 | RGA-M2-2 | Real-coded genetic algorithm with framework M2-2 | | √ | | √ | | | | | | √ | √ | | | | | | |
| 303 | RM-MEDA | Regularity model-based multiobjective estimation of distribution | | √ | | √ | √ | | | | | | | | | | | | |
| 304 | RMOEA/DVA | Robust multi-objective evolutionary algorithm with decision variable assortment | | √ | | √ | √ | | | | | | | | | | | | √ |
| 305 | RMSProp | Root mean square propagation | √ | | | √ | | | | | √ | | | | | | | | |
| 306 | r-NSGA-II | r-dominance based NSGA-II | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 307 | RPD-NSGA-II | Reference point dominance-based NSGA-II | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 308 | RPEA | Reference points-based evolutionary algorithm | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 309 | RSEA | Radial space division based evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 310 | RVEA | Reference vector guided evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 311 | RVEAa | RVEA embedded with the reference vector regeneration strategy | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 312 | RVEA-iGNG | RVEA based on improved growing neural gas | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 313 | S3-CMA-ES | Scalable small subpopulations based covariance matrix adaptation | | √ | √ | √ | √ | | | | √ | | | | | | | | |
| 314 | SA | Simulated annealing | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 315 | SACC-EAM-II | Surrogate-assisted cooperative co-evolutionary algorithm of Minamo | √ | | | √ | √ | | | | | | √ | | | | | | |
| 316 | SACOSO | Surrogate-assisted cooperative swarm optimization | √ | | | √ | √ | | | | √ | | √ | | | | | | |
| 317 | SADE-AMSS | Surrogate-assisted differential evolution with adaptive multi-subspace search | √ | | | √ | √ | | | | | | √ | | | | | | |
| 318 | SADE-ATDSC | Surrogate-assisted differential evolution with adaptation of training data selection criterion | √ | | | √ | √ | | | | | | √ | | | | | | |
| 319 | SADE-Sammon | Sammon mapping assisted differential evolution | √ | | | √ | √ | | | | | | √ | | | | | | |
| 320 | SAMOEA-TL2M | Surrogate-assisted multiobjective evolutionary algorithm based on two-level model management | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 321 | SAMSO | Multiswarm-assisted expensive optimization | √ | | | √ | √ | | | | √ | | √ | | | | | | |
| 322 | SAPO | Surrogate-assisted partial optimization | √ | | | √ | √ | | | | | √ | √ | | | | | | |
| 323 | S-CDAS | Self-controlling dominance area of solutions | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 324 | SCEA | Sparsity clustering basec evolutionary algorithm | | √ | | √ | | | √ | | √ | √ | | | √ | | | | |
| 325 | SCMOEA | Sparse constrained multi-objective evolutionary algorithm | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 326 | SD | Steepest descent | √ | | | √ | | | | | √ | | | | | | | | |
| 327 | S-ECSO | Enhanced competitive swarm optimizer for sparse optimization | | √ | | √ | | | | | √ | | | | √ | | | | |
| 328 | SFA-DE | Scalarization function approximation based differential evolution algorithm | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 329 | SFA-MFDE | Scalarization function approximation based multifactorial differential evolution algorithm | | √ | √ | √ | √ | | | | | | √ | | | | | | |
| 330 | SGEA | Steady-state and generational evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | √ | | | | √ | | | |
| 331 | SGECF | Sparsity-guided elitism co-evolutionary framework | | √ | | √ | | | √ | | √ | √ | | | √ | | | | |
| 332 | SHADE | Success-history based adaptive differential evolution | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 333 | SIBEA | Simple indicator-based evolutionary algorithm | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 334 | SIBEA-kEMOSS | SIBEA with minimum objective subset of size k with minimum error | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 335 | SLMEA | Super-large-scale multi-objective evolutionary algorithm | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 336 | SMEA | Self-organizing multiobjective evolutionary algorithm | | √ | | √ | √ | | | | | | | | | | | | |
| 337 | SMOA | Supervised multi-objective optimization algorithm | | √ | | √ | | | | | | | √ | | | | | | |
| 338 | SMPSO | Speed-constrained multi-objective particle swarm optimization | | √ | | √ | √ | | | | | | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 339 | SMS-EGO | S metric selection based efficient global optimization | | √ | | √ | √ | | | | | | √ | | | | | | |
| 340 | SMS-EMOA | S metric selection based evolutionary multiobjective optimization | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 341 | S-NSGA-II | Sparse NSGA-II | | √ | | √ | | | | | √ | √ | | | √ | | | | |
| 342 | SparseEA | Evolutionary algorithm for sparse multi-objective optimization problems | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 343 | SparseEA2 | Improved SparseEA | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 344 | SparseEMT | Sparse evolutionary multitasking | | √ | | √ | √ | | √ | | √ | √ | | | √ | | | | |
| 345 | SPEA2 | Strength Pareto evolutionary algorithm 2 | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 346 | SPEA2+SDE | SPEA2 with shift-based density estimation | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 347 | SPEA/R | Strength Pareto evolutionary algorithm based on reference direction | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 348 | SQP | Sequential quadratic programming | √ | | | √ | | | | | √ | √ | | | | | | | |
| 349 | SRA | Stochastic ranking algorithm | | | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 350 | SSCEA | Subspace segmentation based co-evolutionary algorithm | | √ | √ | √ | √ | | | | | | | | | | | | |
| 351 | SSDE | Self-organized surrogate-assisted differential evolution | | √ | √ | √ | √ | | | | | √ | √ | | | | | | |
| 352 | SSIO-RL | Search space independent operator based deep reinforcement learning | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 353 | SVR-NSGA-II | Support vector regression based NSGA-II | | √ | | √ | √ | √ | √ | √ | | √ | | | | √ | | | |
| 354 | t-DEA | theta-dominance based evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 355 | tDEA-CPBI | Theta-dominance based evolutionary algorithm with CPBI | | √ | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 356 | TEA | Two-phase evolutionary algorithm | | √ | √ | √ | √ | | | | | √ | √ | | | | | | |
| 357 | TELSO | Two-layer encoding learning swarm optimizer | | √ | | √ | | | √ | | √ | √ | | | √ | | | | |
| 358 | TiGE-2 | Tri-Goal Evolution Framework for CMaOPs | | | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 359 | ToP | Two-phase framework with NSGA-II | | √ | | √ | √ | | | | | √ | | | | | | | |
| 360 | TPCMaO | Three-population based constrained many-objective co-evolutionary algorithm | | | √ | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 361 | TriMOEA-TA&R | Multi-modal MOEA using two-archive and recombination strategies | | √ | | √ | √ | | | | | | | √ | | | | | |
| 362 | TS-NSGA-II | Two-stage NSGA-II | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 363 | TSPP | Two-stage probabilistic penalty | | √ | √ | √ | | | | | | √ | √ | | | | | | |
| 364 | TS-SparseEA | Two-stage SparseEA | | √ | | √ | | | √ | | √ | √ | | | √ | | | | |
| 365 | TSTI | Two-stage evolutionary algorithm with three indicators | | √ | | √ | √ | √ | √ | √ | | √ | | | | | | | |
| 366 | Two_Arch2 | Two-archive algorithm 2 | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 367 | UBCSO | Unbalanced and balanced competition strategy-assisted dual-swarm optimizer | | √ | | √ | | | | | | √ | | | | | | | |
| 368 | UICMO | Utilizing feasible non-dominated solution information for constrained multi-objective optimization | | √ | | √ | | | | | | √ | | | | | | | |

| | 算法缩写 | 算法全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 369 | URCMO | Utilizing the relationship between constrained and unconstrained Pareto fronts for constrained multi-objective optimization | | √ | | √ | √ | | | | | √ | | | | | | | |
| 370 | VaEA | Vector angle based evolutionary algorithm | | √ | √ | √ | √ | √ | √ | √ | | | | | | | | | |
| 371 | WASF-GA | Weighting achievement scalarizing function genetic algorithm | | √ | | √ | √ | √ | √ | √ | | | | | | | | | |
| 372 | WOA | Whale optimization algorithm | √ | | | √ | √ | | | | √ | √ | | | | | | | |
| 373 | WOF | Weighted optimization framework | | √ | | √ | √ | | | | √ | | | | | | | | |
| 374 | WV-MOEA-P | Weight vector based multi-objective optimization algorithm with preference | | √ | | √ | √ | | | | | | | | | | | | |

# 六 问题列表

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | BBOB_F1 | Sphere function | √ | | | √ | | | | | | | √ | | | | | | |
| 2 | BBOB_F2 | Ellipsoidal function | √ | | | √ | | | | | | | √ | | | | | | |
| 3 | BBOB_F3 | Rastrigin function | √ | | | √ | | | | | | | √ | | | | | | |
| 4 | BBOB_F4 | Buche-Rastrigin function | √ | | | √ | | | | | | | √ | | | | | | |
| 5 | BBOB_F5 | Linear slope | √ | | | √ | | | | | | | √ | | | | | | |
| 6 | BBOB_F6 | Attractive sector function | √ | | | √ | | | | | | | √ | | | | | | |
| 7 | BBOB_F7 | Step ellipsoidal function | √ | | | √ | | | | | | | √ | | | | | | |
| 8 | BBOB_F8 | Rosenbrock function | √ | | | √ | | | | | | | √ | | | | | | |
| 9 | BBOB_F9 | Rotated Rosenbrock function | √ | | | √ | | | | | | | √ | | | | | | |
| 10 | BBOB_F10 | Rotated ellipsoidal function | √ | | | √ | | | | | | | √ | | | | | | |
| 11 | BBOB_F11 | Discus function | √ | | | √ | | | | | | | √ | | | | | | |
| 12 | BBOB_F12 | Bent cigar function | √ | | | √ | | | | | | | √ | | | | | | |
| 13 | BBOB_F13 | Sharp ridge function | √ | | | √ | | | | | | | √ | | | | | | |
| 14 | BBOB_F14 | Different powers function | √ | | | √ | | | | | | | √ | | | | | | |
| 15 | BBOB_F15 | Rastrigin function | √ | | | √ | | | | | | | √ | | | | | | |
| 16 | BBOB_F16 | Weierstrass function | √ | | | √ | | | | | | | √ | | | | | | |
| 17 | BBOB_F17 | Schaffers F7 function | √ | | | √ | | | | | | | √ | | | | | | |
| 18 | BBOB_F18 | Moderately ill-conditioned Schaffers F7 function | √ | | | √ | | | | | | | √ | | | | | | |
| 19 | BBOB_F19 | Composite Griewank-Rosenbrock function F8F2 | √ | | | √ | | | | | | | √ | | | | | | |
| 20 | BBOB_F20 | Schwefel function | √ | | | √ | | | | | | | √ | | | | | | |
| 21 | BBOB_F21 | Gallagher's Gaussian 101-me peaks function | √ | | | √ | | | | | | | √ | | | | | | |
| 22 | BBOB_F22 | Gallagher's Gaussian 21-hi peaks function | √ | | | √ | | | | | | | √ | | | | | | |
| 23 | BBOB_F23 | Katsuura function | √ | | | √ | | | | | | | √ | | | | | | |
| 24 | BBOB_F24 | Lunacek bi-Rastrigin function | √ | | | √ | | | | | | | √ | | | | | | |
| 25 | BT1 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |
| 26 | BT2 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |
| 27 | BT3 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |
| 28 | BT4 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |
| 29 | BT5 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |
| 30 | BT6 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |
| 31 | BT7 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |
| 32 | BT8 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |
| 33 | BT9 | Benchmark MOP with bias feature | | √ | | √ | | | | | √ | | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 34 | C10MOP1 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 35 | C10MOP2 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 36 | C10MOP3 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 37 | C10MOP4 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 38 | C10MOP5 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 39 | C10MOP6 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 40 | C10MOP7 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 41 | C10MOP8 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 42 | C10MOP9 | Neural architecture search on CIFAR-10 | | √ | | √ | | | | | √ | | | | | | | | |
| 43 | CEC2008_F1 | Shifted sphere function | √ | | | √ | | | | | √ | | √ | | | | | | |
| 44 | CEC2008_F2 | Shifted Schwefel's function | √ | | | √ | | | | | √ | | √ | | | | | | |
| 45 | CEC2008_F3 | Shifted Rosenbrock's function | √ | | | √ | | | | | √ | | √ | | | | | | |
| 46 | CEC2008_F4 | Shifted Rastrign's function | √ | | | √ | | | | | √ | | √ | | | | | | |
| 47 | CEC2008_F5 | Shifted Griewank's function | √ | | | √ | | | | | √ | | √ | | | | | | |
| 48 | CEC2008_F6 | Shifted Ackley's function | √ | | | √ | | | | | √ | | √ | | | | | | |
| 49 | CEC2008_F7 | FastFractal 'DoubleDip' function | √ | | | √ | | | | | √ | | √ | | | | | | |
| 50 | CEC2010_F1 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 51 | CEC2010_F2 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 52 | CEC2010_F3 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 53 | CEC2010_F4 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 54 | CEC2010_F5 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 55 | CEC2010_F6 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 56 | CEC2010_F7 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 57 | CEC2010_F8 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 58 | CEC2010_F9 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 59 | CEC2010_F10 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 60 | CEC2010_F11 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 61 | CEC2010_F12 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 62 | CEC2010_F13 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 63 | CEC2010_F14 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 64 | CEC2010_F15 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 65 | CEC2010_F16 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 66 | CEC2010_F17 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 67 | CEC2010_F18 | CEC'2010 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 68 | CEC2013_F1 | Shifted elliptic function | √ | | | √ | | | | | √ | | | | | | | | |
| 69 | CEC2013_F2 | Shifted Rastrigin's function | √ | | | √ | | | | | √ | | | | | | | | |
| 70 | CEC2013_F3 | Shifted Ackley's function | √ | | | √ | | | | | √ | | | | | | | | |
| 71 | CEC2013_F4 | 7-nonseparable, 1-separable shifted and rotated elliptic function | √ | | | √ | | | | | √ | | | | | | | | |
| 72 | CEC2013_F5 | 7-nonseparable, 1-separable shifted and rotated Rastrigin's function | √ | | | √ | | | | | √ | | | | | | | | |
| 73 | CEC2013_F6 | 7-nonseparable, 1-separable shifted and rotated Ackley's function | √ | | | √ | | | | | √ | | | | | | | | |
| 74 | CEC2013_F7 | 7-nonseparable, 1-separable shifted and rotated Schwefel's function | √ | | | √ | | | | | √ | | | | | | | | |
| 75 | CEC2013_F8 | 20-nonseparable shifted and rotated elliptic function | √ | | | √ | | | | | √ | | | | | | | | |
| 76 | CEC2013_F9 | 20-nonseparable shifted and rotated Rastrigin's function | √ | | | √ | | | | | √ | | | | | | | | |
| 77 | CEC2013_F10 | 20-nonseparable shifted and rotated Rastrigin's function | √ | | | √ | | | | | √ | | | | | | | | |
| 78 | CEC2013_F11 | 20-nonseparable shifted and rotated Schwefel's function | √ | | | √ | | | | | √ | | | | | | | | |
| 79 | CEC2013_F12 | Shifted Rosenbrock's function | √ | | | √ | | | | | √ | | | | | | | | |
| 80 | CEC2013_F13 | Shifted Schwefel's function with conforming overlapping subcomponents | √ | | | √ | | | | | √ | | | | | | | | |
| 81 | CEC2013_F14 | Shifted Schwefel's function with conflicting overlapping subcomponents | √ | | | √ | | | | | √ | | | | | | | | |
| 82 | CEC2013_F15 | Shifted Schwefel's function | √ | | | √ | | | | | √ | | | | | | | | |
| 83 | CEC2017_F1 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 84 | CEC2017_F2 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 85 | CEC2017_F3 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 86 | CEC2017_F4 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 87 | CEC2017_F5 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 88 | CEC2017_F6 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 89 | CEC2017_F7 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 90 | CEC2017_F8 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 91 | CEC2017_F9 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 92 | CEC2017_F10 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 93 | CEC2017_F11 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 94 | CEC2017_F12 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 95 | CEC2017_F13 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 96 | CEC2017_F14 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 97 | CEC2017_F15 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 98 | CEC2017_F16 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 99 | CEC2017_F17 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 100 | CEC2017_F18 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 101 | CEC2017_F19 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 102 | CEC2017_F20 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 103 | CEC2017_F21 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 104 | CEC2017_F22 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 105 | CEC2017_F23 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 106 | CEC2017_F24 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 107 | CEC2017_F25 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 108 | CEC2017_F26 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 109 | CEC2017_F27 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 110 | CEC2017_F28 | CEC'2017 constrained optimization benchmark problem | √ | | | √ | | | | | | √ | | | | | | | |
| 111 | CEC2020_F1 | Bent cigar function | √ | | | √ | | | | | | | | | | | | | |
| 112 | CEC2020_F2 | Shifted and rotated Schwefel's function | √ | | | √ | | | | | | | | | | | | | |
| 113 | CEC2020_F3 | Shifted and rotated Lunacek bi-Rastrigin function | √ | | | √ | | | | | | | | | | | | | |
| 114 | CEC2020_F4 | Expanded Rosenbrock's plus Griewangk's function | √ | | | √ | | | | | | | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 115 | CEC2020_F5 | Hybrid function 1 | √ | | | √ | | | | | | | | | | | | | |
| 116 | CEC2020_F6 | Hybrid function 2 | √ | | | √ | | | | | | | | | | | | | |
| 117 | CEC2020_F7 | Hybrid function 3 | √ | | | √ | | | | | | | | | | | | | |
| 118 | CEC2020_F8 | Composition function 1 | √ | | | √ | | | | | | | | | | | | | |
| 119 | CEC2020_F9 | Composition function 2 | √ | | | √ | | | | | | | | | | | | | |
| 120 | CEC2020_F10 | Composition function 3 | √ | | | √ | | | | | | | | | | | | | |
| 121 | CF1 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 122 | CF2 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 123 | CF3 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 124 | CF4 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 125 | CF5 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 126 | CF6 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 127 | CF7 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 128 | CF8 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 129 | CF9 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 130 | CF10 | Constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 131 | CI_HS | Multitasking problem (Griewank function + Rastrigin function) | √ | | | √ | | | | | √ | | | | | | √ | | |
| 132 | CI_LS | Multitasking problem (Ackley function + Schwefel function) | √ | | | √ | | | | | √ | | | | | | √ | | |
| 133 | CI_MS | Multitasking problem (Ackley function + Rastrigin function) | √ | | | √ | | | | | √ | | | | | | √ | | |
| 134 | CitySegMOP1 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 135 | CitySegMOP2 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 136 | CitySegMOP3 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 137 | CitySegMOP4 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 138 | CitySegMOP5 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 139 | CitySegMOP6 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 140 | CitySegMOP7 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 141 | CitySegMOP8 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 142 | CitySegMOP9 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 143 | CitySegMOP10 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 144 | CitySegMOP11 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 145 | CitySegMOP12 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 146 | CitySegMOP13 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 147 | CitySegMOP14 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 148 | CitySegMOP15 | Neural architecture search on Cityscape segmentation datasets | | √ | | √ | | | | | √ | | √ | | | | | | |
| 149 | Community Detection | The community detection problem with label based encoding | √ | | | | | √ | | | √ | | √ | | | | | | |
| 150 | DAS-CMOP1 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 151 | DAS-CMOP2 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 152 | DAS-CMOP3 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 153 | DAS-CMOP4 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 154 | DAS-CMOP5 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 155 | DAS-CMOP6 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 156 | DAS-CMOP7 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 157 | DAS-CMOP8 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 158 | DAS-CMOP9 | Difficulty-adjustable and scalable constrained benchmark MOP | | √ | | √ | | | | | √ | √ | | | | | | | |
| 159 | DOC1 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 160 | DOC2 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 161 | DOC3 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 162 | DOC4 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 163 | DOC5 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 164 | DOC6 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 165 | DOC7 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 166 | DOC8 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 167 | DOC9 | Benchmark MOP with constraints in decision and objective spaces | | √ | | √ | | | | | | √ | | | | | | | |
| 168 | DSMOP1 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 169 | DSMOP2 | Dynamic sparse multi-objective optimization | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | problem | | | | | | | | | | | | | | | | | |
| 170 | DSMOP3 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 171 | DSMOP4 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 172 | DSMOP5 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 173 | DSMOP6 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 174 | DSMOP7 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 175 | DSMOP8 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 176 | DSMOP9 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 177 | DSMOP10 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 178 | DSMOP11 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 179 | DSMOP12 | Dynamic sparse multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | | √ | √ | | | |
| 180 | DTLZ1 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 181 | DTLZ2 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 182 | DTLZ3 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 183 | DTLZ4 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 184 | DTLZ5 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 185 | DTLZ6 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 186 | DTLZ7 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 187 | DTLZ8 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 188 | DTLZ9 | Benchmark MOP proposed by Deb, Thiele, Laumanns, and Zitzler | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 189 | CDTLZ2 | Convex DTLZ2 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 190 | IDTLZ1 | Inverted DTLZ1 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 191 | IDTLZ2 | Inverted DTLZ2 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 192 | IDTLZ3 | Inverted DTLZ3 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 193 | IDTLZ4 | Inverted DTLZ4 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 194 | Minus-DTLZ1 | Minus DTLZ1 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 195 | Minus-DTLZ2 | Minus DTLZ2 | | √ | √ | √ | | | | | √ | | √ | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 196 | Minus-DTLZ3 | Minus DTLZ3 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 197 | Minus-DTLZ4 | Minus DTLZ4 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 198 | Minus-DTLZ5 | Minus DTLZ5 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 199 | Minus-DTLZ6 | Minus DTLZ6 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 200 | Minus-DTLZ7 | Minus DTLZ7 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 201 | Minus-IDTLZ1 | Minus IDTLZ1 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 202 | Minus-IDTLZ2 | Minus IDTLZ2 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 203 | Minus-IDTLZ3 | Minus IDTLZ3 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 204 | Minus-IDTLZ4 | Minus IDTLZ4 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 205 | SDTLZ1 | Scaled DTLZ1 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 206 | SDTLZ2 | Scaled DTLZ2 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 207 | C1-DTLZ1 | Constrained DTLZ1 | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 208 | C1-DTLZ3 | Constrained DTLZ3 | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 209 | C2-DTLZ2 | Constrained DTLZ2 | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 210 | C3-DTLZ4 | Constrained DTLZ4 | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 211 | DC1-DTLZ1 | DTLZ1 with constrains in decision space | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 212 | DC1-DTLZ3 | DTLZ3 with constrains in decision space | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 213 | DC2-DTLZ1 | DTLZ1 with constrains in decision space | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 214 | DC2-DTLZ3 | DTLZ3 with constrains in decision space | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 215 | DC3-DTLZ1 | DTLZ1 with constrains in decision space | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 216 | DC3-DTLZ3 | DTLZ3 with constrains in decision space | | √ | √ | √ | | | | | √ | √ | √ | | | | | | |
| 217 | EOPCCV_F1 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 218 | EOPCCV_F2 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 219 | EOPCCV_F3 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 220 | EOPCCV_F4 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 221 | EOPCCV_F5 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 222 | EOPCCV_F6 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 223 | EOPCCV_F7 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 224 | EOPCCV_F8 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 225 | EOPCCV_F9 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 226 | EOPCCV_F10 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 227 | EOPCCV_F11 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 228 | EOPCCV_F12 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 229 | EOPCCV_F13 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 230 | EOPCCV_F14 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 231 | EOPCCV_F15 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 232 | EOPCCV_F16 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 233 | EOPCCV_F17 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 234 | EOPCCV_F18 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 235 | EOPCCV_F19 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 236 | EOPCCV_F20 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 237 | EOPCCV_F21 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 238 | EOPCCV_F22 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 239 | EOPCCV_F23 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 240 | EOPCCV_F24 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 241 | EOPCCV_F25 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 242 | EOPCCV_F26 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 243 | EOPCCV_F27 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 244 | EOPCCV_F28 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 245 | EOPCCV_F29 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 246 | EOPCCV_F30 | Expensive optimization problems with continuous and categorical variables | √ | | | √ | | √ | | | | | √ | | | | | | |
| 247 | FCP1 | Benchmark constrained MOP proposed by Yuan | | √ | | √ | | | | | | √ | | | | | | | |
| 248 | FCP2 | Benchmark constrained MOP proposed by Yuan | | √ | | √ | | | | | | √ | | | | | | | |
| 249 | FCP3 | Benchmark constrained MOP proposed by Yuan | | √ | | √ | | | | | | √ | | | | | | | |
| 250 | FCP4 | Benchmark constrained MOP proposed by Yuan | | √ | | √ | | | | | | √ | | | | | | | |
| 251 | FCP5 | Benchmark constrained MOP proposed by Yuan | | √ | | √ | | | | | | √ | | | | | | | |
| 252 | FDA1 | Benchmark dynamic MOP proposed by Farina, Deb, and Amato | | √ | | √ | | | | | √ | | | | | √ | | | |
| 253 | FDA2 | Benchmark dynamic MOP proposed by Farina, Deb, and Amato | | √ | | √ | | | | | √ | | | | | √ | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 254 | FDA3 | Benchmark dynamic MOP proposed by Farina, Deb, and Amato | | √ | | √ | | | | | √ | | | | | √ | | | |
| 255 | FDA4 | Benchmark dynamic MOP proposed by Farina, Deb, and Amato | | √ | | √ | | | | | √ | | | | | √ | | | |
| 256 | FDA5 | Benchmark dynamic MOP proposed by Farina, Deb, and Amato | | √ | | √ | | | | | √ | | | | | √ | | | |
| 257 | GLSMOP1 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 258 | GLSMOP2 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 259 | GLSMOP3 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 260 | GLSMOP4 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 261 | GLSMOP5 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 262 | GLSMOP6 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 263 | GLSMOP7 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 264 | GLSMOP8 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 265 | GLSMOP9 | General large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 266 | HYL1 | Multi-modal multi-objective benchmark problem proposed by Hisao, Yiming, and Lie Meng | | √ | | √ | | | | | | | | √ | | | | | |
| 267 | HYL2 | Multi-modal multi-objective benchmark problem proposed by Hisao, Yiming, and Lie Meng | | √ | | √ | | | | | | | | √ | | | | | |
| 268 | HYL3 | Multi-modal multi-objective benchmark problem proposed by Hisao, Yiming, and Lie Meng | | √ | | √ | | | | | | | | √ | | | | | |
| 269 | HYL4 | Multi-modal multi-objective benchmark problem proposed by Hisao, Yiming, and Lie Meng | | √ | | √ | | | | | | | | √ | | | | | |
| 270 | HYL5 | Multi-modal multi-objective benchmark problem proposed by Hisao, Yiming, and Lie Meng | | √ | | √ | | | | | | | | √ | | | | | |
| 271 | IMMOEA_F1 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 272 | IMMOEA_F2 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 273 | IMMOEA_F3 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 274 | IMMOEA_F4 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 275 | IMMOEA_F5 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 276 | IMMOEA_F6 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 277 | IMMOEA_F7 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 278 | IMMOEA_F8 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 279 | IMMOEA_F9 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 280 | IMMOEA_F10 | Benchmark MOP for testing IM-MOEA | | √ | | √ | | | | | √ | | | | | | | | |
| 281 | IMOP1 | Benchmark MOP with irregular Pareto front | | √ | | √ | | | | | | | √ | | | | | | |
| 282 | IMOP2 | Benchmark MOP with irregular Pareto front | | √ | | √ | | | | | | | √ | | | | | | |
| 283 | IMOP3 | Benchmark MOP with irregular Pareto front | | √ | | √ | | | | | | | √ | | | | | | |
| 284 | IMOP4 | Benchmark MOP with irregular Pareto front | | √ | | √ | | | | | | | √ | | | | | | |
| 285 | IMOP5 | Benchmark MOP with irregular Pareto front | | √ | | √ | | | | | | | √ | | | | | | |
| 286 | IMOP6 | Benchmark MOP with irregular Pareto front | | √ | | √ | | | | | | | √ | | | | | | |
| 287 | IMOP7 | Benchmark MOP with irregular Pareto front | | √ | | √ | | | | | | | √ | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 288 | IMOP8 | Benchmark MOP with irregular Pareto front | | √ | | √ | | | | | | | √ | | | | | | |
| 289 | IN1KMOP1 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 290 | IN1KMOP2 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 291 | IN1KMOP3 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 292 | IN1KMOP4 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 293 | IN1KMOP5 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 294 | IN1KMOP6 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 295 | IN1KMOP7 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 296 | IN1KMOP8 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 297 | IN1KMOP9 | Neural architecture search on ImageNet 1K | | √ | | √ | | | | | √ | | √ | | | | | | |
| 298 | Instance1 | Multitasking multi-objective problem (ZDT4-R + ZDT4-G) | | √ | | √ | | | | | √ | | | | | | √ | | |
| 299 | Instance2 | Multitasking multi-objective problem (ZDT4-RC + ZDT4-A) | | √ | | √ | | | | | √ | √ | | | | | √ | | |
| 300 | KP | The knapsack problem | √ | | | | | | √ | | √ | √ | | | | | | | |
| 301 | LIR-CMOP1 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 302 | LIR-CMOP2 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 303 | LIR-CMOP3 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 304 | LIR-CMOP4 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 305 | LIR-CMOP5 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 306 | LIR-CMOP6 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 307 | LIR-CMOP7 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 308 | LIR-CMOP8 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 309 | LIR-CMOP9 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 310 | LIR-CMOP10 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 311 | LIR-CMOP11 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 312 | LIR-CMOP12 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 313 | LIR-CMOP13 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 314 | LIR-CMOP14 | Constrained benchmark MOP with large infeasible regions | | √ | | √ | | | | | √ | √ | | | | | | | |
| 315 | LRMOP1 | Large-scale robust multi-objective benchmark problem | | √ | √ | √ | | | | | √ | | √ | | √ | | | | √ |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 316 | LRMOP2 | Large-scale robust multi-objective benchmark problem | | √ | √ | √ | | | | | √ | | √ | | √ | | | | √ |
| 317 | LRMOP3 | Large-scale robust multi-objective benchmark problem | | √ | √ | √ | | | | | √ | | √ | | √ | | | | √ |
| 318 | LRMOP4 | Large-scale robust multi-objective benchmark problem | | √ | √ | √ | | | | | √ | | √ | | √ | | | | √ |
| 319 | LRMOP5 | Large-scale robust multi-objective benchmark problem | | √ | √ | √ | | | | | √ | | √ | | √ | | | | √ |
| 320 | LRMOP6 | Large-scale robust multi-objective benchmark problem | | √ | √ | √ | | | | | √ | | √ | | √ | | | | √ |
| 321 | LSCM1 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 322 | LSCM2 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 323 | LSCM3 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 324 | LSCM4 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 325 | LSCM5 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 326 | LSCM6 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 327 | LSCM7 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 328 | LSCM8 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 329 | LSCM9 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 330 | LSCM10 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 331 | LSCM11 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 332 | LSCM12 | Large-scale constrained multiobjective benchmark problem | | √ | | √ | | | | | √ | √ | | | | | | | |
| 333 | LSMOP1 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 334 | LSMOP2 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 335 | LSMOP3 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 336 | LSMOP4 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 337 | LSMOP5 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 338 | LSMOP6 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 339 | LSMOP7 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 340 | LSMOP8 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 341 | LSMOP9 | Large-scale benchmark MOP | | √ | √ | √ | | | | | √ | | | | | | | | |
| 342 | MaF1 | Inverted DTLZ1 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 343 | MaF2 | DTLZ2BZ | | √ | √ | √ | | | | | √ | | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 344 | MaF3 | Convex DTLZ3 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 345 | MaF4 | Inverted and scaled DTLZ3 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 346 | MaF5 | Scaled DTLZ4 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 347 | MaF6 | DTLZ5IM | | √ | √ | √ | | | | | √ | | | | | | | | |
| 348 | MaF7 | DTLZ7 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 349 | MaF8 | MP-DMP | | √ | √ | √ | | | | | | | | | | | | | |
| 350 | MaF9 | ML-DMP | | √ | √ | √ | | | | | | | | | | | | | |
| 351 | MaF10 | WFG1 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 352 | MaF11 | WFG2 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 353 | MaF12 | WFG9 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 354 | MaF13 | P7 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 355 | MaF14 | LSMOP3 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 356 | MaF15 | Inverted LSMOP8 | | √ | √ | √ | | | | | √ | | | | | | | | |
| 357 | MaOPP_binary | Many-objective pathfinding problem based on binary encoding | | | √ | | | | √ | | √ | | √ | | | | | | |
| 358 | MaOPP_real | Many-objective pathfinding problem based on real encoding | | | √ | √ | | | | | √ | | √ | | | | | | |
| 359 | Mario | Play with Mario | √ | | | | √ | √ | | | | | | | | | | | |
| 360 | MaxCut | The max-cut problem | √ | | | | | | √ | | √ | | | | | | | | |
| 361 | MLDMP | The multi-line distance minimization problem | | √ | √ | √ | | | | | | | | | | | | | |
| 362 | MMF1 | Multi-modal multi-objective test function | | √ | | √ | | | | | | | | √ | | | | | |
| 363 | MMF2 | Multi-modal multi-objective test function | | √ | | √ | | | | | | | | √ | | | | | |
| 364 | MMF3 | Multi-modal multi-objective test function | | √ | | √ | | | | | | | | √ | | | | | |
| 365 | MMF4 | Multi-modal multi-objective test function | | √ | | √ | | | | | | | | √ | | | | | |
| 366 | MMF5 | Multi-modal multi-objective test function | | √ | | √ | | | | | | | | √ | | | | | |
| 367 | MMF6 | Multi-modal multi-objective test function | | √ | | √ | | | | | | | | √ | | | | | |
| 368 | MMF7 | Multi-modal multi-objective test function | | √ | | √ | | | | | | | | √ | | | | | |
| 369 | MMF8 | Multi-modal multi-objective test function | | √ | | √ | | | | | | | | √ | | | | | |
| 370 | MMMOP1 | Multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | | | | √ | | | | | |
| 371 | MMMOP2 | Multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | | | | √ | | | | | |
| 372 | MMMOP3 | Multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | | | | √ | | | | | |
| 373 | MMMOP4 | Multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | | | | √ | | | | | |
| 374 | MMMOP5 | Multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | | | | √ | | | | | |
| 375 | MMMOP6 | Multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | | | | √ | | | | | |
| 376 | MMOP_HS1 | Large-scale sparse multitasking multi-objective optimization problem | | √ | | √ | | | | | √ | | | | √ | | √ | | |
| 377 | MMOP_HS2 | Large-scale sparse multitasking multi-objective optimization problem | | √ | | √ | | | | | √ | | | | √ | | √ | | |
| 378 | MMOP_LS1 | Large-scale sparse multitasking multi-objective optimization problem | | √ | | √ | | | | | √ | | | | √ | | √ | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 379 | MMOP_LS2 | Large-scale sparse multitasking multi-objective optimization problem | | √ | | √ | | | | | √ | | | | √ | | √ | | |
| 380 | MMOP_MS1 | Large-scale sparse multitasking multi-objective optimization problem | | √ | | √ | | | | | √ | | | | √ | | √ | | |
| 381 | MMOP_MS2 | Large-scale sparse multitasking multi-objective optimization problem | | √ | | √ | | | | | √ | | | | √ | | √ | | |
| 382 | MMOP_NS1 | Large-scale sparse multitasking multi-objective optimization problem | | √ | | √ | | | | | √ | | | | √ | | √ | | |
| 383 | MMOP_NS2 | Large-scale sparse multitasking multi-objective optimization problem | | √ | | √ | | | | | √ | | | | √ | | √ | | |
| 384 | MOEADDE_F1 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 385 | MOEADDE_F2 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 386 | MOEADDE_F3 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 387 | MOEADDE_F4 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 388 | MOEADDE_F5 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 389 | MOEADDE_F6 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 390 | MOEADDE_F7 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 391 | MOEADDE_F8 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 392 | MOEADDE_F9 | Benchmark MOP for testing MOEA/D-DE | | √ | | √ | | | | | √ | | | | | | | | |
| 393 | MOEADM2M_F1 | Benchmark MOP for testing MOEA/D-M2M | | √ | | √ | | | | | √ | | | | | | | | |
| 394 | MOEADM2M_F2 | Benchmark MOP for testing MOEA/D-M2M | | √ | | √ | | | | | √ | | | | | | | | |
| 395 | MOEADM2M_F3 | Benchmark MOP for testing MOEA/D-M2M | | √ | | √ | | | | | √ | | | | | | | | |
| 396 | MOEADM2M_F4 | Benchmark MOP for testing MOEA/D-M2M | | √ | | √ | | | | | √ | | | | | | | | |
| 397 | MOEADM2M_F5 | Benchmark MOP for testing MOEA/D-M2M | | √ | | √ | | | | | √ | | | | | | | | |
| 398 | MOEADM2M_F6 | Benchmark MOP for testing MOEA/D-M2M | | √ | | √ | | | | | √ | | | | | | | | |
| 399 | MOEADM2M_F7 | Benchmark MOP for testing MOEA/D-M2M | | √ | | √ | | | | | √ | | | | | | | | |
| 400 | MOKP | The multi-objective knapsack problem | | √ | √ | | | | √ | | √ | √ | | | | | | | |
| 401 | MONRP | The multi-objective next release problem | | √ | | | | | √ | | √ | | | | | | | | |
| 402 | MOTSP | The multi-objective traveling salesman problem | | √ | √ | | | | | √ | √ | | | | | | | | |
| 403 | MPDMP | The multi-point distance minimization problem | | √ | √ | √ | | | | | | | | | | | | | |
| 404 | mQAP | The multi-objective quadratic assignment problem | | √ | √ | | | | | √ | √ | | | | | | | | |
| 405 | MW1 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 406 | MW2 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 407 | MW3 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 408 | MW4 | Constrained benchmark MOP proposed by Ma and Wang | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 409 | MW5 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 410 | MW6 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 411 | MW7 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 412 | MW8 | Constrained benchmark MOP proposed by Ma and Wang | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 413 | MW9 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 414 | MW10 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 415 | MW11 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 416 | MW12 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 417 | MW13 | Constrained benchmark MOP proposed by Ma and Wang | | √ | | √ | | | | | √ | √ | | | | | | | |
| 418 | MW14 | Constrained benchmark MOP proposed by Ma and Wang | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 419 | NI_HS | Multitasking problem (Rosenbrock function + Rastrigin function) | √ | | | √ | | | | | √ | | | | | | √ | | |
| 420 | NI_MS | Multitasking problem (Griewank function + Weierstrass function) | √ | | | √ | | | | | √ | | | | | | √ | | |
| 421 | RMMEDA_F1 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 422 | RMMEDA_F2 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 423 | RMMEDA_F3 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 424 | RMMEDA_F4 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 425 | RMMEDA_F5 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 426 | RMMEDA_F6 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 427 | RMMEDA_F7 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 428 | RMMEDA_F8 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 429 | RMMEDA_F9 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 430 | RMMEDA_F10 | Benchmark MOP for testing RM-MEDA | | √ | | √ | | | | | √ | | | | | | | | |
| 431 | RWMOP1 | Pressure vessal problem | | √ | | √ | | | | | | √ | | | | | | | |
| 432 | RWMOP2 | Vibrating platform | | √ | | √ | | | | | | √ | | | | | | | |
| 433 | RWMOP3 | Two bar truss design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 434 | RWMOP4 | Weldan beam design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 435 | RWMOP5 | Disc brake design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 436 | RWMOP6 | Speed reducer design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 437 | RWMOP7 | Gear train design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 438 | RWMOP8 | Car side impact design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 439 | RWMOP9 | Four bar plane truss | | √ | | √ | | | | | | √ | | | | | | | |
| 440 | RWMOP10 | Two bar plane truss | | √ | | √ | | | | | | √ | | | | | | | |
| 441 | RWMOP11 | Water resource management problem | | √ | | √ | | | | | | √ | | | | | | | |
| 442 | RWMOP12 | Simply supported I-beam design | | √ | | √ | | | | | | √ | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 443 | RWMOP13 | Gear box design | | √ | | √ | | | | | | √ | | | | | | | |
| 444 | RWMOP14 | Multiple-disk clutch brake design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 445 | RWMOP15 | Spring design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 446 | RWMOP16 | Cantilever beam design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 447 | RWMOP17 | Bulk carriers design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 448 | RWMOP18 | Front rail design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 449 | RWMOP19 | Multi-product batch plant | | √ | | √ | | | | | | √ | | | | | | | |
| 450 | RWMOP20 | Hydro-static thrust bearing design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 451 | RWMOP21 | Crash energy management for high-speed train | | √ | | √ | | | | | | √ | | | | | | | |
| 452 | RWMOP22 | Haverly's pooling problem | | √ | | √ | | | | | | √ | | | | | | | |
| 453 | RWMOP23 | Reactor network design | | √ | | √ | | | | | | √ | | | | | | | |
| 454 | RWMOP24 | Heat exchanger network design | | √ | | √ | | | | | | √ | | | | | | | |
| 455 | RWMOP25 | Process synthesis problem | | √ | | √ | | | | | | √ | | | | | | | |
| 456 | RWMOP26 | Process sythesis and design problem | | √ | | √ | | | | | | √ | | | | | | | |
| 457 | RWMOP27 | Process flow sheeting problem | | √ | | √ | | | | | | √ | | | | | | | |
| 458 | RWMOP28 | Two reactor problem | | √ | | √ | | | | | | √ | | | | | | | |
| 459 | RWMOP29 | Process synthesis problem | | √ | | √ | | | | | | √ | | | | | | | |
| 460 | RWMOP30 | Synchronous pptimal pulse-width modulation of 3-level inverters | | √ | | √ | | | | | | √ | | | | | | | |
| 461 | RWMOP31 | Synchronous pptimal pulse-width modulation of 5-level inverters | | √ | | √ | | | | | | √ | | | | | | | |
| 462 | RWMOP32 | Synchronous pptimal pulse-width modulation of 7-level inverters | | √ | | √ | | | | | | √ | | | | | | | |
| 463 | RWMOP33 | Synchronous pptimal pulse-width modulation of 9-level inverters | | √ | | √ | | | | | | √ | | | | | | | |
| 464 | RWMOP34 | Synchronous pptimal pulse-width modulation of 11-level inverters | | √ | | √ | | | | | | √ | | | | | | | |
| 465 | RWMOP35 | Synchronous pptimal pulse-width modulation of 13-level inverters | | √ | | √ | | | | | | √ | | | | | | | |
| 466 | RWMOP36 | Optimal sizing of single phase distributed generation with reactive power support for phase balancing at main transformer/grid and active power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 467 | RWMOP37 | Optimal Sizing of Single Phase Distributed Generation with reactive power support for Phase Balancing at Main Transformer/Grid and reactive Power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 468 | RWMOP38 | Optimal sizing of single phase distributed generation with reactive power support for active and reactive power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 469 | RWMOP39 | Optimal sizing of single phase distributed generation with reactive power support for phase balancing at main transformer/grid and active and reactive power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 470 | RWMOP40 | Optimal power flow for minimizing active and reactive power loss | | √ | | √ | | | | | | √ | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| 471 | RWMOP41 | Optimal power flow for minimizing voltage deviation, active and reactive power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 472 | RWMOP42 | Optimal power flow for minimizing voltage deviation, and active power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 473 | RWMOP43 | Optimal power flow for minimizing fuel cost, and active power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 474 | RWMOP44 | Optimal power flow for minimizing fuel cost, active and reactive power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 475 | RWMOP45 | Optimal power flow for minimizing fuel cost, voltage deviation, and active power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 476 | RWMOP46 | Optimal power flow for minimizing fuel cost, voltage deviation, active and reactive power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 477 | RWMOP47 | Optimal droop setting for minimizing active and reactive power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 478 | RWMOP48 | Optimal droop setting for minimizing voltage deviation and active power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 479 | RWMOP49 | Optimal droop setting for minimizing voltage deviation, active, and reactive power loss | | √ | | √ | | | | | | √ | | | | | | | |
| 480 | RWMOP50 | Power distribution system planning | | √ | | √ | | | | | | √ | | | | | | | |
| 481 | SCMOP1 | Benchmark CMOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | √ | √ | | √ | | | | |
| 482 | SCMOP2 | Benchmark CMOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | √ | √ | | √ | | | | |
| 483 | SCMOP3 | Benchmark CMOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | √ | √ | | √ | | | | |
| 484 | SCMOP4 | Benchmark CMOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | √ | √ | | √ | | | | |
| 485 | SCMOP5 | Benchmark CMOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | √ | √ | | √ | | | | |
| 486 | SCMOP6 | Benchmark CMOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | √ | √ | | √ | | | | |
| 487 | SCMOP7 | Benchmark CMOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | √ | √ | | √ | | | | |
| 488 | SCMOP8 | Benchmark CMOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | √ | √ | | √ | | | | |
| 489 | SDC1 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 490 | SDC2 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 491 | SDC3 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 492 | SDC4 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 493 | SDC5 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 494 | SDC6 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 495 | SDC7 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| 496 | SDC8 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 497 | SDC9 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 498 | SDC10 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 499 | SDC11 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 500 | SDC12 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 501 | SDC13 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 502 | SDC14 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 503 | SDC15 | Scalable high-dimensional decicsion constraint benchamrk | | √ | | √ | | | | | | √ | | | | | | | |
| 504 | SMD1 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | | | | | | | √ | |
| 505 | SMD2 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | | | | | | | √ | |
| 506 | SMD3 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | | | | | | | √ | |
| 507 | SMD4 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | | | | | | | √ | |
| 508 | SMD5 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | | | | | | | √ | |
| 509 | SMD6 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | | | | | | | √ | |
| 510 | SMD7 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | | | | | | | √ | |
| 511 | SMD8 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | | | | | | | √ | |
| 512 | SMD9 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | √ | | | | | | √ | |
| 513 | SMD10 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | √ | | | | | | √ | |
| 514 | SMD11 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | √ | | | | | | √ | |
| 515 | SMD12 | Bilevel optimization problems proposed by Sinha, Malo, and Deb | | √ | | √ | | | | | | √ | | | | | | √ | |
| 516 | SO_ISCSO_2016 | International student competition in structural optimization | √ | | | | √ | | | | √ | √ | | | | | | | |
| 517 | SO_ISCSO_2017 | International student competition in structural optimization | √ | | | | √ | | | | √ | √ | | | | | | | |
| 518 | SO_ISCSO_2018 | International student competition in structural optimization | √ | | | | √ | | | | √ | √ | | | | | | | |
| 519 | SO_ISCSO_2019 | International student competition in structural optimization | √ | | | | √ | | | | √ | √ | | | | | | | |
| 520 | SO_ISCSO_2021 | International student competition in structural | √ | | | | √ | | | | √ | √ | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| | | optimization | | | | | | | | | | | | | | | | | |
| 521 | SO_ISCSO_2022 | International student competition in structural optimization | √ | | | | √ | | | | √ | √ | | | | | | | |
| 522 | Sparse_CD | The community detection problem | | √ | | | | | √ | | √ | | √ | | √ | | | | |
| 523 | Sparse_CN | The critical node detection problem | | √ | | | | | √ | | √ | | √ | | √ | | | | |
| 524 | Sparse_FS | The feature selection problem | | √ | | | | | √ | | √ | | √ | | √ | | | | |
| 525 | Sparse_IS | The instance selection problem | | √ | | | | | √ | | √ | | √ | | √ | | | | |
| 526 | Sparse_KP | The sparse multi-objective knapsack problem | | √ | √ | | | | √ | | √ | | | | | | | | |
| 527 | Sparse_NN | The neural network training problem | | √ | | √ | | | | | √ | | √ | | √ | | | | |
| 528 | Sparse_PM | The pattern mining problem | | √ | | | | | √ | | √ | | √ | | √ | | | | |
| 529 | Sparse_PO | The portfolio optimization problem | | √ | | √ | | | | | √ | | √ | | √ | | | | |
| 530 | Sparse_SR | The sparse signal reconstruction problem | | √ | | √ | | | | | √ | | √ | | √ | | | | |
| 531 | SMMOP1 | Sparse multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | √ | √ | | | | |
| 532 | SMMOP2 | Sparse multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | √ | √ | | | | |
| 533 | SMMOP3 | Sparse multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | √ | √ | | | | |
| 534 | SMMOP4 | Sparse multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | √ | √ | | | | |
| 535 | SMMOP5 | Sparse multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | √ | √ | | | | |
| 536 | SMMOP6 | Sparse multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | √ | √ | | | | |
| 537 | SMMOP7 | Sparse multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | √ | √ | | | | |
| 538 | SMMOP8 | Sparse multi-modal multi-objective optimization problem | | √ | √ | √ | | | | | √ | | | √ | √ | | | | |
| 539 | SMOP1 | Benchmark MOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | | √ | | √ | | | | |
| 540 | SMOP2 | Benchmark MOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | | √ | | √ | | | | |
| 541 | SMOP3 | Benchmark MOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | | √ | | √ | | | | |
| 542 | SMOP4 | Benchmark MOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | | √ | | √ | | | | |
| 543 | SMOP5 | Benchmark MOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | | √ | | √ | | | | |
| 544 | SMOP6 | Benchmark MOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | | √ | | √ | | | | |
| 545 | SMOP7 | Benchmark MOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | | √ | | √ | | | | |
| 546 | SMOP8 | Benchmark MOP with sparse Pareto optimal solutions | | √ | √ | √ | | | | | √ | | √ | | √ | | | | |
| 547 | SOP_F1 | Sphere function | √ | | | √ | | | | | | | √ | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
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| 548 | SOP_F2 | Schwefel's function 2.22 | √ | | | √ | | | | | | | √ | | | | | | |
| 549 | SOP_F3 | Schwefel's function 1.2 | √ | | | √ | | | | | | | √ | | | | | | |
| 550 | SOP_F4 | Schwefel's function 2.21 | √ | | | √ | | | | | | | √ | | | | | | |
| 551 | SOP_F5 | Generalized Rosenbrock's function | √ | | | √ | | | | | | | √ | | | | | | |
| 552 | SOP_F6 | Step function | √ | | | √ | | | | | | | √ | | | | | | |
| 553 | SOP_F7 | Quartic function with noise | √ | | | √ | | | | | | | √ | | | | | | |
| 554 | SOP_F8 | Generalized Schwefel's function 2.26 | √ | | | √ | | | | | | | √ | | | | | | |
| 555 | SOP_F9 | Generalized Rastrigin's function | √ | | | √ | | | | | | | √ | | | | | | |
| 556 | SOP_F10 | Ackley's function | √ | | | √ | | | | | | | √ | | | | | | |
| 557 | SOP_F11 | Generalized Griewank's function | √ | | | √ | | | | | | | √ | | | | | | |
| 558 | SOP_F12 | Generalized penalized function | √ | | | √ | | | | | | | √ | | | | | | |
| 559 | SOP_F13 | Generalized penalized function | √ | | | √ | | | | | | | √ | | | | | | |
| 560 | SOP_F14 | Shekel's foxholes function | √ | | | √ | | | | | | | √ | | | | | | |
| 561 | SOP_F15 | Kowalik's function | √ | | | √ | | | | | | | √ | | | | | | |
| 562 | SOP_F16 | Six-hump camel-back function | √ | | | √ | | | | | | | √ | | | | | | |
| 563 | SOP_F17 | Branin function | √ | | | √ | | | | | | | √ | | | | | | |
| 564 | SOP_F18 | Goldstein-price function | √ | | | √ | | | | | | | √ | | | | | | |
| 565 | SOP_F19 | Hartman's family | √ | | | √ | | | | | | | √ | | | | | | |
| 566 | SOP_F20 | Hartman's family | √ | | | √ | | | | | | | √ | | | | | | |
| 567 | SOP_F21 | Shekel's family | √ | | | √ | | | | | | | √ | | | | | | |
| 568 | SOP_F22 | Shekel's family | √ | | | √ | | | | | | | √ | | | | | | |
| 569 | SOP_F23 | Shekel's family | √ | | | √ | | | | | | | √ | | | | | | |
| 570 | TP1 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 571 | TP2 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 572 | TP3 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 573 | TP4 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 574 | TP5 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 575 | TP6 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 576 | TP7 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 577 | TP8 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 578 | TP9 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | | | | | | | | √ |
| 579 | TP10 | Test problem for robust multi-objective optimization | | √ | | √ | | | | | √ | √ | | | | | | | √ |
| 580 | TREE1 | The time-varying ratio error estimation problem | | √ | | √ | | | | | √ | √ | √ | | | | | | |
| 581 | TREE2 | The time-varying ratio error estimation problem | | √ | | √ | | | | | √ | √ | √ | | | | | | |
| 582 | TREE3 | The time-varying ratio error estimation problem | | √ | | √ | | | | | √ | √ | √ | | | | | | |
| 583 | TREE4 | The time-varying ratio error estimation problem | | √ | | √ | | | | | √ | √ | √ | | | | | | |
| 584 | TREE5 | The time-varying ratio error estimation problem | | √ | | √ | | | | | √ | √ | √ | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 585 | TREE6 | The time-varying ratio error estimation problem | | √ | | √ | | | | | √ | √ | √ | | | | | | |
| 586 | TSP | The traveling salesman problem | √ | | | | | | | √ | √ | | | | | | | | |
| 587 | UF1 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 588 | UF2 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 589 | UF3 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 590 | UF4 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 591 | UF5 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 592 | UF6 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 593 | UF7 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 594 | UF8 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 595 | UF9 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 596 | UF10 | Unconstrained benchmark MOP | | √ | | √ | | | | | √ | | | | | | | | |
| 597 | VNT1 | Benchmark MOP proposed by Viennet | | √ | | √ | | | | | | | | | | | | | |
| 598 | VNT2 | Benchmark MOP proposed by Viennet | | √ | | √ | | | | | | | | | | | | | |
| 599 | VNT3 | Benchmark MOP proposed by Viennet | | √ | | √ | | | | | | | | | | | | | |
| 600 | VNT4 | Benchmark MOP proposed by Viennet | | √ | | √ | | | | | | √ | | | | | | | |
| 601 | WFG1 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 602 | WFG2 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 603 | WFG3 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 604 | WFG4 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 605 | WFG5 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 606 | WFG6 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 607 | WFG7 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 608 | WFG8 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 609 | WFG9 | Benchmark MOP proposed by Walking Fish Group | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 610 | Minus-WFG1 | Minus WFG1 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 611 | Minus-WFG2 | Minus WFG2 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 612 | Minus-WFG3 | Minus WFG3 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 613 | Minus-WFG4 | Minus WFG4 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 614 | Minus-WFG5 | Minus WFG5 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 615 | Minus-WFG6 | Minus WFG6 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 616 | Minus-WFG7 | Minus WFG7 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 617 | Minus-WFG8 | Minus WFG8 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 618 | Minus-WFG9 | Minus WFG9 | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 619 | ZCAT1 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 620 | ZCAT2 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 621 | ZCA3 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 622 | ZCA4 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 623 | ZCA5 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 624 | ZCAT6 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 625 | ZCAT7 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 626 | ZCAT8 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 627 | ZCAT9 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 628 | ZCAT10 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 629 | ZCAT11 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 630 | ZCAT12 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 631 | ZCAT13 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 632 | ZCAT14 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 633 | ZCAT15 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 634 | ZCAT16 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 635 | ZCAT17 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 636 | ZCAT18 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 637 | ZCAT19 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 638 | ZCAT20 | Benchmark MOP proposed by Zapotecas, Coello, Aguirre, and Tanaka | | √ | √ | √ | | | | | √ | | √ | | | | | | |
| 639 | ZDT1 | Benchmark MOP proposed by Zitzler, Deb, and Thiele | | √ | | √ | | | | | √ | | √ | | | | | | |
| 640 | ZDT2 | Benchmark MOP proposed by Zitzler, Deb, and Thiele | | √ | | √ | | | | | √ | | √ | | | | | | |
| 641 | ZDT3 | Benchmark MOP proposed by Zitzler, Deb, and Thiele | | √ | | √ | | | | | √ | | √ | | | | | | |
| 642 | ZDT4 | Benchmark MOP proposed by Zitzler, Deb, and Thiele | | √ | | √ | | | | | √ | | √ | | | | | | |
| 643 | ZDT5 | Benchmark MOP proposed by Zitzler, Deb, and Thiele | | √ | | | | | √ | | √ | | √ | | | | | | |
| 644 | ZDT6 | Benchmark MOP proposed by Zitzler, Deb, and Thiele | | √ | | √ | | | | | √ | | √ | | | | | | |
| 645 | ZXH_CF1 | Constrained benchmark MOP proposed by | | √ | √ | √ | | | | | √ | √ | | | | | | | |

| | 问题缩写 | 问题全称 | single | multi | many | real | integer | label | binary | permutation | large | constrained | expensive | multimodal | sparse | dynamic | multitask | bilevel | robust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Zhou, Xiang, and He | | | | | | | | | | | | | | | | | |
| 646 | ZXH_CF2 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 647 | ZXH_CF3 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 648 | ZXH_CF4 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 649 | ZXH_CF5 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 650 | ZXH_CF6 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 651 | ZXH_CF7 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 652 | ZXH_CF8 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 653 | ZXH_CF9 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 654 | ZXH_CF10 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 655 | ZXH_CF11 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 656 | ZXH_CF12 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 657 | ZXH_CF13 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 658 | ZXH_CF14 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 659 | ZXH_CF15 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |
| 660 | ZXH_CF16 | Constrained benchmark MOP proposed by Zhou, Xiang, and He | | √ | √ | √ | | | | | √ | √ | | | | | | | |